

Spleen_mass_analysis

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First time run order: Spleen → CBC → Spleen

This markdown produces clean_meta_data.csv used by CBC and ELISA analysis.

The section looking at the correlation between spleen size and lymphocyte counts

at the bottom requires cleaned_combined_cbc_data.csv from CBC. That's why CBC needs to be ran on the second try. But, this is only applicable if this is the first time this code is ran.

Load the needed library first

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.1      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(readxl)
library(writexl)
library(janitor)
```

```
##
## Attaching package: 'janitor'
##
## The following objects are masked from 'package:stats':
##
##   chisq.test, fisher.test
```

```
library(factoextra)
```

```
## Welcome! Want to learn more? See two factoextra-related books at https://goo.gl/ve3WBa
```

```
library(skimr)
library(naniar)
```

```
##
## Attaching package: 'naniar'
##
## The following object is masked from 'package:skimr':
##
##     n_complete
```

```
library(ggplot2)
library(ggpubr)
library(emmeans)
```

```
## Welcome to emmeans.
## Caution: You lose important information if you filter this package's results.
## See '? untidy'
```

```
library(lme4)
```

```
## Loading required package: Matrix
##
## Attaching package: 'Matrix'
##
## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack
```

```
library(lmerTest)
```

```
##
## Attaching package: 'lmerTest'
##
## The following object is masked from 'package:lme4':
##
##     lmer
##
## The following object is masked from 'package:stats':
##
##     step
```

```
library(scales)
```

```
##
## Attaching package: 'scales'
```

```
##
## The following object is masked from 'package:purrr':
##
##   discard
##
## The following object is masked from 'package:readr':
##
##   col_factor
```

```
library(DHARMA)
```

```
## This is DHARMA 0.4.7. For overview type '?DHARMA'. For recent changes, type news(package = 'DHARMA')
```

Setting the paths

```
base_path <- "/Users/zehao/Documents/Research Projects/Graham Group Thesis Research (Summer 2025)"

data_path <- file.path(base_path,
                        "Metadata")
clean_data_path <- file.path(base_path,
                              "Cleaned data")
figure_path <- file.path(base_path,
                          "Figures")
spleen_figure_path <- file.path(figure_path,
                                 "Spleen Figures")
dir.create(spleen_figure_path,
           showWarnings = FALSE,
           recursive = TRUE)
```

Plot saver

```
plot_saver <- function(filename,
                        plot,
                        path = spleen_figure_path,
                        width = 6,
                        height = 4,
                        dpi = 300) {
  ggsave(filename = filename,
          plot = plot,
          path = path,
          width = width,
          height = height,
          units = "in",
          dpi = dpi)
}
```

Loading the data

```
spleen_data <- read_excel(file.path(data_path,
                                   "SF Project Spleen Data (2025).xlsx"),
                          sheet = "spleen cell count data")
body_mass_data <- read_excel(file.path(data_path,
                                       "metamouse25_12_16_25_ver.xlsx"),
                             sheet = "mass and fecal data")
spleen_lym_count <- read_excel(file.path(data_path,
                                         "SF Project Spleen Data (2025).xlsx"),
                              sheet = "spleen cell concentration calcu")
meta_data <- read_excel(file.path(data_path,
                                  "metamouse25_12_16_25_ver.xlsx"),
                        sheet = "metadata")

cbc_data <- read_csv(file.path(clean_data_path,
                               "cleaned_combined_cbc_data.csv"))

## Rows: 324 Columns: 20
## -- Column specification -----
## Delimiter: ","
## chr (8): cage_wedge, strain, experiment, t_muris_dose, lab_or_rw, cohort, s...
## dbl (12): mouse_id, neu_number, lym_number, mon_number, eos_number, bas_numb...
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
```

Setting the colors (Okabe-Ito palette and Paul Tol's Bright)

```
color_black <- "#000000"
color_mustard <- "#E69F00"
color_light_blue <- "#56B4E9"
color_green <- "#009E73"
color_yellow <- "#F0E442"
color_dark_blue <- "#0072B2"
color_orange <- "#D55E00"
color_purple <- "#CC79A7"

color_grey <- "#BBBBBB"
color_dark_purple <- "#AA3377"
color_light_purple <- "#EE6677"
color_sand <- "#CCBB44"
color_cyan <- "#66CCEE"
color_olive <- "#228833"
color_navy <- "#4477AA"
```

Cleaning the spleen data

```
clean_spleen_data <- spleen_data %>%
  filter(experiment == "NSF",
         !is.na(`spleen mass (g)`) ) %>%
  select(mouse.id,
         strain,
         T.muris.dose,
         lab.or.rw,
         `spleen mass (g)`) %>%
  clean_names() %>%
  mutate(mouse_id = str_trim(as.character(mouse_id)),
         strain = as.character(strain),
         t_muris_dose = case_match(t_muris_dose, "0" ~ "Zero", .default = t_muris_dose),
         t_muris_dose = factor(t_muris_dose,
                               levels = c("Zero", "Low", "High")),
         lab_or_rw = factor(lab_or_rw),
         spleen_mass_g = as.numeric(spleen_mass_g))

str(clean_spleen_data)
```

```
## tibble [105 x 5] (S3: tbl_df/tbl/data.frame)
## $ mouse_id      : chr [1:105] "1301" "1302" "1303" "1304" ...
## $ strain        : chr [1:105] "B10.BR" "B10.BR" "B10.BR" "B10.BR" ...
## $ t_muris_dose  : Factor w/ 3 levels "Zero","Low","High": 1 2 3 1 3 2 2 3 1 3 ...
## $ lab_or_rw     : Factor w/ 2 levels "Lab","RW": 2 2 2 1 2 2 1 1 2 2 ...
## $ spleen_mass_g: num [1:105] 0.1062 0.0987 0.1135 0.0548 0.1189 ...
```

```
# Only using the NSF ones since I only have spleens from those
```

Cleaning the mass data

```
clean_body_mass_data <- body_mass_data %>%
  filter(week == 3 | week == 4,
         !is.na(mass)) %>%
  select(mouse.id,
         mass,
         `cage/wedge`) %>%
  clean_names() %>%
  mutate(mouse_id = str_trim(as.character(mouse_id)),
         mass = as.numeric(mass),
         cage_wedge = if_else(cage_wedge == "spare",
                              "25",
                              cage_wedge),
         cage_wedge = factor(cage_wedge))

str(clean_body_mass_data)
```

```
## tibble [109 x 3] (S3: tbl_df/tbl/data.frame)
```

```
## $ mouse_id : chr [1:109] "1304" "1305" "1309" "1310" ...
## $ mass      : num [1:109] 22.8 25 23.7 23.3 22.3 23.2 23.3 24.4 23.7 21.8 ...
## $ cage_wedge: Factor w/ 29 levels "1","10","11",...: 1 1 12 12 19 19 20 20 21 21 ...
```

```
# Let's assign the spares to cage 25 for ease of calculations.
# I'm going to only use week 3 and 4 since that's the end of the experiment
```

Cleaning splenocyte data

```
clean_spleen_lym_count <- spleen_lym_count %>%
  clean_names() %>%
  select(avg_total_cell_concentration_cell_m_l,
         solution_volume_m_l,
         mouse_id) %>%
  mutate(mouse_id = as.character(mouse_id) %>% str_trim(),
         total_lym_number = avg_total_cell_concentration_cell_m_l * solution_volume_m_l,
         total_lym_number = as.numeric(total_lym_number)) %>%
  select(mouse_id,
         total_lym_number)
```

Cleaning meta data

```
clean_meta_data <- meta_data %>%
  clean_names() %>%
  filter(experiment == "NSF",
         !is.na(t_muris_dose)) %>%
  select(mouse_id,
         strain,
         experiment,
         t_muris_dose,
         lab_or_rw,
         cohort,
         source) %>%
  mutate(mouse_id = str_trim(as.character(mouse_id)),
         strain = as.character(strain),
         t_muris_dose = case_match(t_muris_dose, "0" ~ "Zero",
                                   .default = t_muris_dose),
         t_muris_dose = factor(t_muris_dose,
                               levels = c("Zero",
                                           "Low",
                                           "High")),
         lab_or_rw = factor(lab_or_rw),
         experiment = as.character(experiment))

str(clean_meta_data)
```

```
## tibble [110 x 7] (S3: tbl_df/tbl/data.frame)
## $ mouse_id : chr [1:110] "1301" "1302" "1303" "1304" ...
```

```
## $ strain      : chr [1:110] "B10.BR" "B10.BR" "B10.BR" "B10.BR" ...
## $ experiment  : chr [1:110] "NSF" "NSF" "NSF" "NSF" ...
## $ t_muris_dose: Factor w/ 3 levels "Zero","Low","High": 1 2 3 1 3 3 2 2 3 1 ...
## $ lab_or_rw   : Factor w/ 2 levels "Lab","RW": 2 2 2 1 1 2 2 1 1 2 ...
## $ cohort      : chr [1:110] "NSF1" "NSF1" "NSF1" "NSF1" ...
## $ source      : chr [1:110] "JAX" "JAX" "JAX" "JAX" ...
```

```
write_csv(clean_meta_data,
          file.path(clean_data_path,
                    "clean_meta_data.csv"))
```

We want to include in a file called “compiled data.” But, before that I want to check to make sure the strains, ID, and the lab/rw conditions all match across the two data sets.

1. Mouse ID
2. Strain
3. NSF Experiment
4. T. muris Dosage
5. Lab or RW
6. Week 3 Body Mass
7. Week 3 Spleen Mass

Checking for mistakes

```
compiled_data_strain_check <- clean_meta_data %>%
  full_join(clean_spleen_data,
            by = "mouse_id") %>%
  mutate(match = strain.x == strain.y) %>%
  filter(!match)
```

```
compiled_data_strain_check
```

```
## # A tibble: 0 x 12
## # i 12 variables: mouse_id <chr>, strain.x <chr>, experiment <chr>,
## #   t_muris_dose.x <fct>, lab_or_rw.x <fct>, cohort <chr>, source <chr>,
## #   strain.y <chr>, t_muris_dose.y <fct>, lab_or_rw.y <fct>,
## #   spleen_mass_g <dbl>, match <lgl>
```

```
compiled_data_lab_rw_check <- clean_meta_data %>%
  full_join(clean_spleen_data,
            by = "mouse_id") %>%
  mutate(match = lab_or_rw.x == lab_or_rw.y) %>%
  filter(!match)
```

```
compiled_data_lab_rw_check
```

```
## # A tibble: 0 x 12
## # i 12 variables: mouse_id <chr>, strain.x <chr>, experiment <chr>,
## #   t_muris_dose.x <fct>, lab_or_rw.x <fct>, cohort <chr>, source <chr>,
## #   strain.y <chr>, t_muris_dose.y <fct>, lab_or_rw.y <fct>,
## #   spleen_mass_g <dbl>, match <lgl>
```

```
compiled_data_t_muris_dose_check <- clean_meta_data %>%
  full_join(clean_spleen_data,
            by = "mouse_id") %>%
  mutate(match = t_muris_dose.x == t_muris_dose.y) %>%
  filter(!match)
```

```
compiled_data_t_muris_dose_check
```

```
## # A tibble: 0 x 12
## # i 12 variables: mouse_id <chr>, strain.x <chr>, experiment <chr>,
## #   t_muris_dose.x <fct>, lab_or_rw.x <fct>, cohort <chr>, source <chr>,
## #   strain.y <chr>, t_muris_dose.y <fct>, lab_or_rw.y <fct>,
## #   spleen_mass_g <dbl>, match <lgl>
```

#Mouse 1350 has different worm dose depending on the sheet, inner join will not include that data

Creating compiled data

```
spleen_mass_compiled_data <- clean_meta_data %>%
  left_join(clean_spleen_data,
            by = "mouse_id") %>%
  left_join(clean_body_mass_data,
            by = "mouse_id") %>%
  filter(!is.na(spleen_mass_g),
         !is.na(mass)) %>%
  select(mouse_id,
         strain = strain.x,
         lab_or_rw = lab_or_rw.x,
         t_muris_dose = t_muris_dose.x,
         spleen_mass_g,
         mass,
         cage_wedge,
         source,
         cohort) %>%
  mutate(strain = factor(strain),
         strain = relevel(strain,
                          ref = "C57BL/6J"),
         t_muris_dose = factor(t_muris_dose,
                               levels = c("Zero",
                                           "Low",
                                           "High")),
         lab_or_rw = factor(lab_or_rw,
                             levels = c("Lab",
                                         "RW"))))
```

Now I will calculate the spleen-mass ratio (SMR) and make a plot. Still thinking whether this is the correct way to do this. See the last plot.

Calculating spleen-mass ratio (SMR)

```
spleen_mass_compiled_data <- spleen_mass_compiled_data %>%
  mutate(spleen_mass_ratio = spleen_mass_g/mass)

lym_number_spleen_data <- spleen_mass_compiled_data %>%
  left_join(clean_spleen_lym_count,
            by = "mouse_id") %>%
  filter(!is.na(total_lym_number))

write_csv(lym_number_spleen_data,
          file.path(clean_data_path,
                    "lym_number_spleen_data.csv"))

write_csv(spleen_mass_compiled_data,
          file.path(clean_data_path,
                    "clean_compiled_spleen_data.csv"))
```

Filte the cbc data to join with spleen data

```
cbc_lym <- cbc_data %>%
  filter(phase == "Post-RW") %>%
  select(mouse_id,
         lym_number) %>%
  mutate(mouse_id = str_trim(as.character(mouse_id)))

spleen_cbc <- spleen_mass_compiled_data %>%
  left_join(cbc_lym,
            by = "mouse_id") %>%
  filter(!is.na(lym_number),
         !is.na(spleen_mass_g))
```

Plotting SMR data

```
spleen_mass_ratio_plot <- ggplot(spleen_mass_compiled_data,
                                aes(x = t_muris_dose,
                                    y = spleen_mass_ratio,
                                    fill = lab_or_rw,
                                    color = lab_or_rw,
                                    group = interaction(t_muris_dose,
                                                         lab_or_rw))) +
  geom_boxplot(position = position_dodge(width = 0.75),
              alpha = 0.8,
              outlier.shape = NA,
              width = 0.5,
              color = "black",
              linewidth = 0.6) +
```

```

geom_jitter(position = position_jitterdodge(jitter.width = 0.15,
                                             dodge.width = 0.75),

            alpha = 0.8,
            size = 1.8,
            shape = 21,
            color = "black") +
facet_wrap(~ strain) +
scale_fill_manual(values = c("RW" = color_green,
                             "Lab" = color_mustard),
                  name = expression(bold("Environment")),
                  labels = c("Lab" = "Laboratory",
                             "RW" = "Rewilded")) +
scale_color_manual(values = c("RW" = color_green,
                              "Lab" = color_mustard),
                   name = expression(bold("Environment")),
                   labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
scale_y_continuous(name = expression(bold("Spleen to Body Mass Ratio (SMR)")),
                   expand = expansion(mult = c(0.05, 0.3))) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),

      strip.text = element_text(size = 10,
                                face = "bold"),

      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),

      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
xlab(expression(bolditalic("T. muris")~bold("Dose")) +
ggtitle("Spleen to Body Mass Ratio (SMR)")

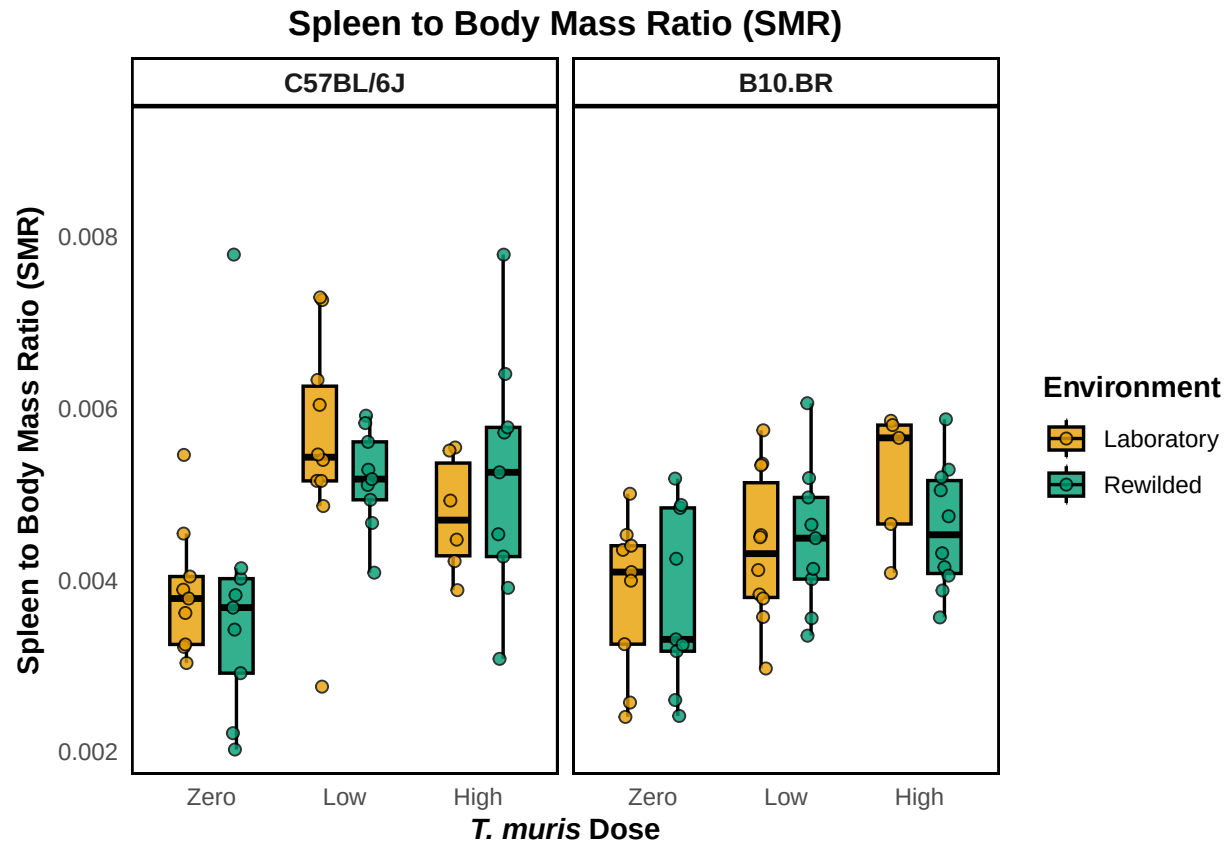
spleen_mass_ratio_plot

```

```

## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.

```



```
plot_saver("smr_general.png",
  spleen_mass_ratio_plot)
```

```
## Warning: No shared levels found between 'names(values)' of the manual scale and the
## data's colour values.
```

lmer modeling for SMR

```
smr_lmer <- lmer(spleen_mass_ratio ~
  strain *
  lab_or_rw *
  t_muris_dose +
  (1 | cage_wedge),
  data = spleen_mass_compiled_data)
```

```
summary(smr_lmer)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: spleen_mass_ratio ~ strain * lab_or_rw * t_muris_dose + (1 |
##   cage_wedge)
## Data: spleen_mass_compiled_data
```

```

##
## REML criterion at convergence: -980.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.7135 -0.5489 -0.0281  0.5031  3.9909
##
## Random effects:
##   Groups      Name      Variance Std.Dev.
## cage_wedge (Intercept) 9.416e-08 0.0003069
## Residual              9.861e-07 0.0009930
## Number of obs: 104, groups: cage_wedge, 29
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)      3.883e-03  3.490e-04  8.502e+01
## strainB10.BR     -4.405e-05  4.915e-04  8.684e+01
## lab_or_rwRW      -9.859e-05  5.279e-04  4.644e+01
## t_muris_doseLow   1.666e-03  4.676e-04  9.184e+01
## t_muris_doseHigh  8.851e-04  5.415e-04  9.168e+01
## strainB10.BR:lab_or_rwRW  6.504e-05  7.452e-04  4.719e+01
## strainB10.BR:t_muris_doseLow -1.138e-03  6.569e-04  9.083e+01
## strainB10.BR:t_muris_doseHigh  5.029e-04  7.866e-04  9.190e+01
## lab_or_rwRW:t_muris_doseLow -2.675e-04  6.617e-04  8.931e+01
## lab_or_rwRW:t_muris_doseHigh  5.182e-04  7.165e-04  9.123e+01
## strainB10.BR:lab_or_rwRW:t_muris_doseLow  3.805e-04  9.331e-04  8.822e+01
## strainB10.BR:lab_or_rwRW:t_muris_doseHigh -1.102e-03  1.023e-03  9.105e+01
##
##              t value Pr(>|t|)
## (Intercept)      11.124 < 2e-16 ***
## strainB10.BR      -0.090  0.928784
## lab_or_rwRW       -0.187  0.852671
## t_muris_doseLow    3.562  0.000586 ***
## t_muris_doseHigh   1.634  0.105606
## strainB10.BR:lab_or_rwRW  0.087  0.930825
## strainB10.BR:t_muris_doseLow -1.733  0.086498 .
## strainB10.BR:t_muris_doseHigh  0.639  0.524213
## lab_or_rwRW:t_muris_doseLow -0.404  0.687001
## lab_or_rwRW:t_muris_doseHigh  0.723  0.471414
## strainB10.BR:lab_or_rwRW:t_muris_doseLow  0.408  0.684457
## strainB10.BR:lab_or_rwRW:t_muris_doseHigh -1.077  0.284195
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##
##      (Intr) stB10.BR lb__RW t_mr_L t_mr_H stB10.BR:__RW sB10.BR:__L
## stranB10.BR      -0.710
## lab_or_rwRW      -0.661  0.470
## t_murs_dsLw      -0.713  0.506  0.471
## t_mrs_dsHgh      -0.624  0.443  0.413  0.470
## stB10.BR:__RW     0.468 -0.659 -0.708 -0.334 -0.292
## sB10.BR:__L       0.508 -0.708 -0.336 -0.712 -0.334  0.467
## sB10.BR:__H       0.430 -0.603 -0.284 -0.323 -0.688  0.398  0.454
## lb_r_RW:__L       0.504 -0.358 -0.647 -0.707 -0.332  0.458  0.503
## lb_r_RW:__H       0.472 -0.335 -0.603 -0.355 -0.756  0.427  0.253

```

```
## sB10.BR:__RW:__L -0.357  0.499    0.459  0.501  0.235 -0.645      -0.704
## sB10.BR:__RW:__H -0.330  0.464    0.422  0.249  0.529 -0.594      -0.349
##               sB10.BR:__H 1__RW:__L 1__RW:__H sB10.BR:__RW:__L
## stranB10.BR
## lab_or_rwRW
## t_murs_dsLw
## t_mrs_dsHgh
## stB10.BR:__RW
## sB10.BR:__L
## sB10.BR:__H
## lb_r_RW:__L      0.229
## lb_r_RW:__H      0.520      0.482
## sB10.BR:__RW:__L -0.319      -0.709      -0.342
## sB10.BR:__RW:__H -0.769      -0.337      -0.700      0.475
```

```
emm_smr <- emmeans(smr_lmer,
                    specs = c("t_muris_dose", "lab_or_rw"),
                    by = c("strain"),
                    type = "response")
emm_smr
```

```
## strain = C57BL/6J:
## t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
## Zero         Lab      0.00388 0.000352 75.8  0.00318  0.00458
## Low          Lab      0.00555 0.000330 83.3  0.00489  0.00620
## High         Lab      0.00477 0.000427 85.3  0.00392  0.00562
## Zero         RW       0.00378 0.000396 12.9  0.00293  0.00464
## Low          RW       0.00518 0.000396 12.9  0.00433  0.00604
## High         RW       0.00519 0.000396 12.9  0.00433  0.00604
##
## strain = B10.BR:
## t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
## Zero         Lab      0.00384 0.000347 83.3  0.00315  0.00453
## Low          Lab      0.00437 0.000329 82.5  0.00371  0.00502
## High         Lab      0.00523 0.000468 85.9  0.00430  0.00616
## Zero         RW       0.00381 0.000396 12.9  0.00295  0.00466
## Low          RW       0.00445 0.000396 12.9  0.00359  0.00530
## High         RW       0.00461 0.000382 11.1  0.00377  0.00545
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
pairs(emm_smr)
```

```
## strain = C57BL/6J:
## contrast      estimate      SE    df t.ratio p.value
## Zero Lab - Low Lab -1.67e-03 0.000472 91.6  -3.528  0.0084
## Zero Lab - High Lab -8.85e-04 0.000548 91.2  -1.614  0.5914
## Zero Lab - Zero RW   9.86e-05 0.000530 26.0   0.186  1.0000
## Zero Lab - Low RW  -1.30e-03 0.000530 26.0  -2.452  0.1757
## Zero Lab - High RW  -1.30e-03 0.000530 26.0  -2.462  0.1726
## Low Lab - High Lab   7.81e-04 0.000528 90.3   1.478  0.6789
## Low Lab - Zero RW    1.76e-03 0.000516 25.9   3.422  0.0227
```

```
## Low Lab - Low RW      3.66e-04 0.000516 25.9    0.710 0.9790
## Low Lab - High RW     3.61e-04 0.000516 25.9    0.700 0.9803
## High Lab - Zero RW    9.84e-04 0.000583 34.0    1.689 0.5482
## High Lab - Low RW    -4.14e-04 0.000583 34.0   -0.711 0.9792
## High Lab - High RW   -4.20e-04 0.000583 34.0   -0.720 0.9781
## Zero RW - Low RW     -1.40e-03 0.000468 74.2   -2.987 0.0426
## Zero RW - High RW    -1.40e-03 0.000470 75.3   -2.984 0.0428
## Low RW - High RW     -5.15e-06 0.000470 75.3   -0.011 1.0000
##
## strain = B10.BR:
## contrast      estimate      SE    df t.ratio p.value
## Zero Lab - Low Lab -5.27e-04 0.000463 84.4  -1.138 0.8642
## Zero Lab - High Lab -1.39e-03 0.000577 92.0  -2.405 0.1652
## Zero Lab - Zero RW   3.36e-05 0.000527 27.2   0.064 1.0000
## Zero Lab - Low RW    -6.07e-04 0.000527 27.2  -1.151 0.8555
## Zero Lab - High RW   -7.70e-04 0.000516 25.3  -1.492 0.6722
## Low Lab - High Lab   -8.61e-04 0.000559 89.6  -1.539 0.6400
## Low Lab - Zero RW     5.61e-04 0.000515 25.7   1.088 0.8816
## Low Lab - Low RW     -7.94e-05 0.000515 25.7  -0.154 1.0000
## Low Lab - High RW    -2.43e-04 0.000504 23.8  -0.482 0.9964
## High Lab - Zero RW    1.42e-03 0.000613 37.8   2.318 0.2122
## High Lab - Low RW     7.81e-04 0.000613 37.8   1.274 0.7970
## High Lab - High RW    6.18e-04 0.000604 36.0   1.023 0.9070
## Zero RW - Low RW     -6.40e-04 0.000470 75.3  -1.361 0.7495
## Zero RW - High RW    -8.04e-04 0.000457 74.5  -1.759 0.4980
## Low RW - High RW     -1.63e-04 0.000457 74.5  -0.358 0.9992
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
emm_smr_df <- as.data.frame(emm_smr)

emm_smr_plot <- ggplot(emm_smr_df) +
  geom_point(aes(x = t_muris_dose,
                 y = emmean,
                 fill = lab_or_rw,
                 group = lab_or_rw),
             position = position_dodge(width = 0.5),
             alpha = 1,
             size = 3,
             shape = 21,
             color = "black") +
  geom_errorbar(aes(x = t_muris_dose,
                    ymin = lower.CL,
                    ymax = upper.CL,
                    color = lab_or_rw,
                    group = lab_or_rw),
               width = 0.3,
               alpha = 0.8,
               position = position_dodge(width = 0.5)) +
  geom_point(data = spleen_mass_compiled_data,
             aes(x = t_muris_dose,
                 y = spleen_mass_ratio,
                 color = lab_or_rw),
```

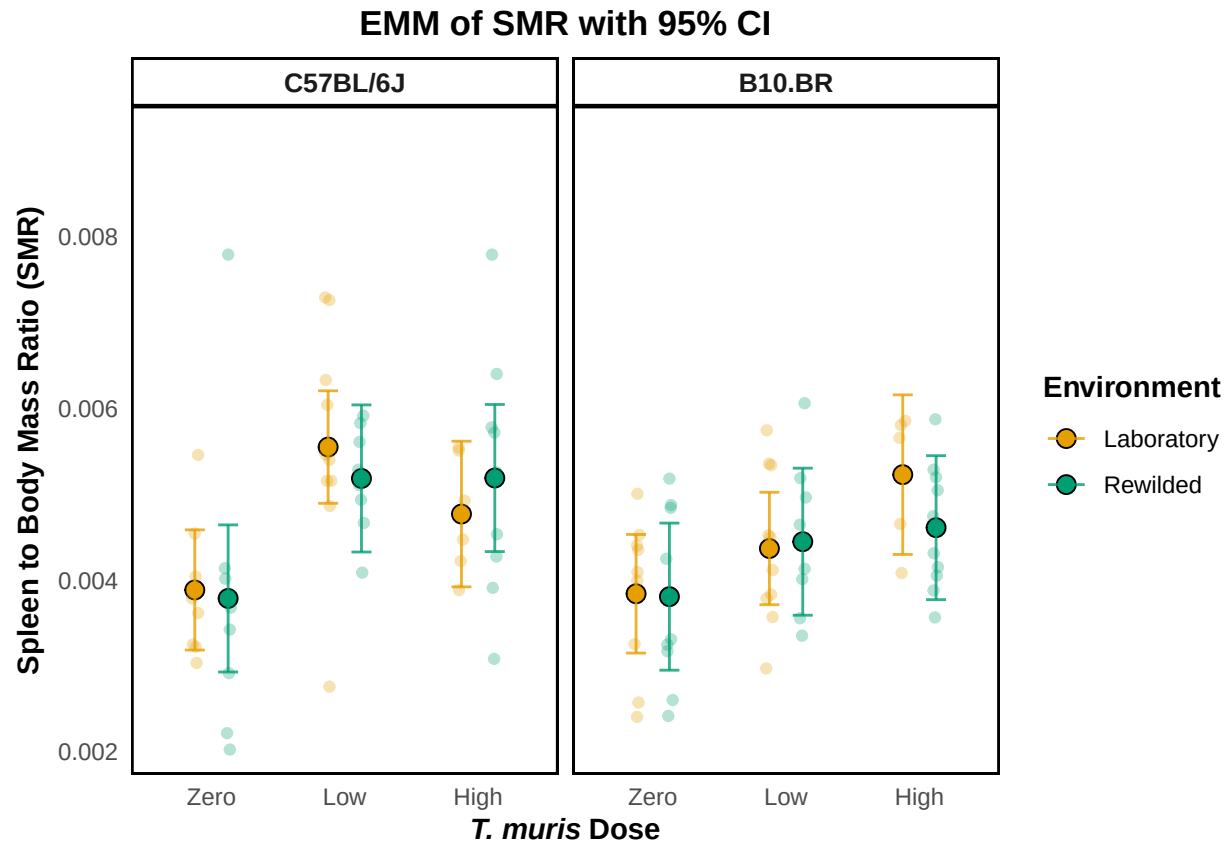
```

    position = position_jitterdodge(jitter.width = 0.1,
                                    dodge.width = 0.5),

    alpha = 0.3,
    size = 1.5) +
scale_y_continuous(name = expression(bold("Spleen to Body Mass Ratio (SMR)")),
                   expand = expansion(mult = c(0.05, 0.3))) +
scale_x_discrete(name = expression(bolditalic("T. muris")~bold("Dose")) +
scale_fill_manual(values = c("Lab" = color_mustard,
                             "RW" = color_green),
                  name = expression(bold("Environment")),
                  labels = c("Lab" = "Laboratory",
                             "RW" = "Rewilded")) +
scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                   name = expression(bold("Environment")),
                   labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
facet_wrap(~ strain) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle("EMM of SMR with 95% CI")

emm_smr_plot

```

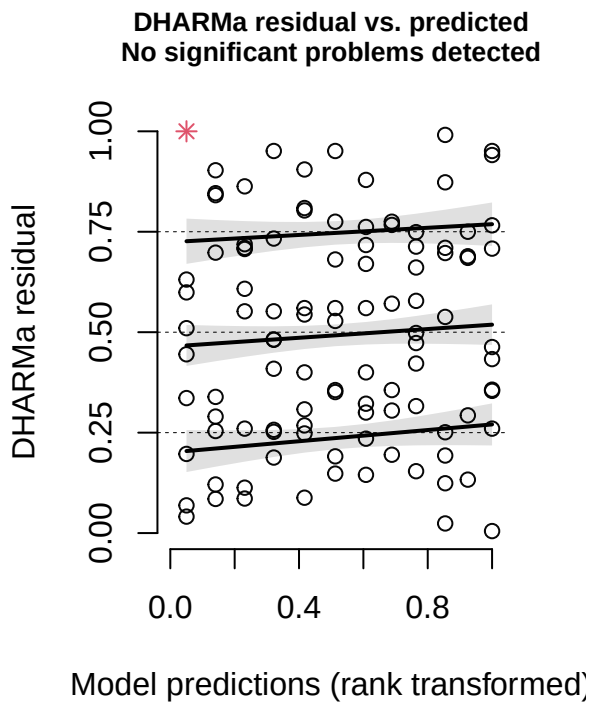
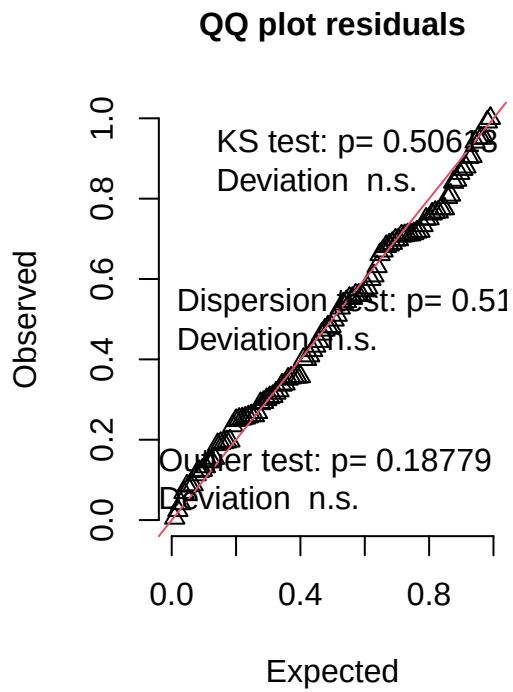


```
plot_saver("emm_smr_plot.png",
  emm_smr_plot)
```

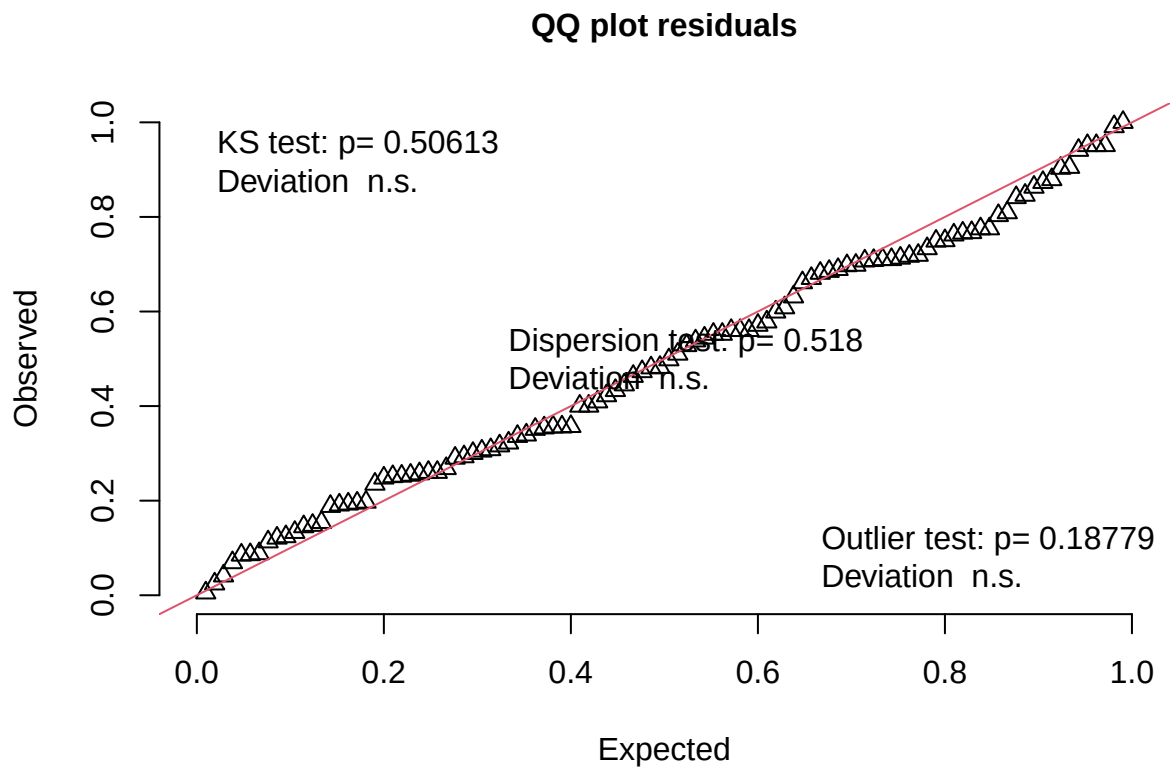
```
smr_sim_res <- simulateResiduals(fittedModel = smr_lmer, n = 1000)
```

```
plot(smr_sim_res)
```


DHARMa residual



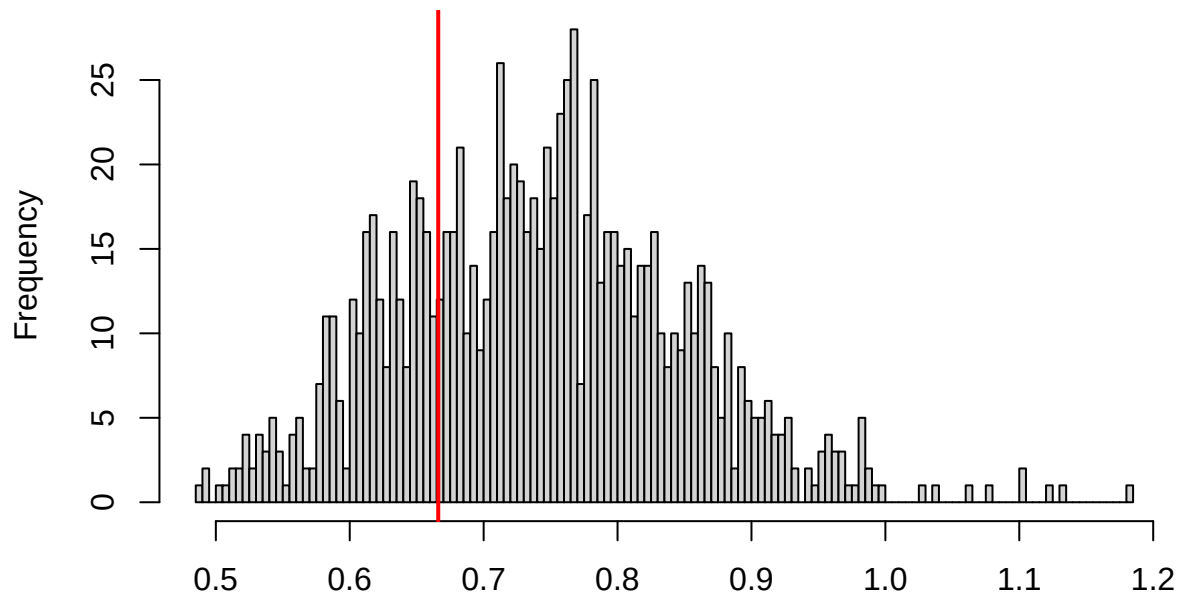
```
testUniformity(smr_sim_res)
```



```
##  
## Asymptotic one-sample Kolmogorov-Smirnov test  
##  
## data: simulationOutput$scaledResiduals  
## D = 0.080769, p-value = 0.5061  
## alternative hypothesis: two-sided
```

```
testDispersion(smr_sim_res)
```

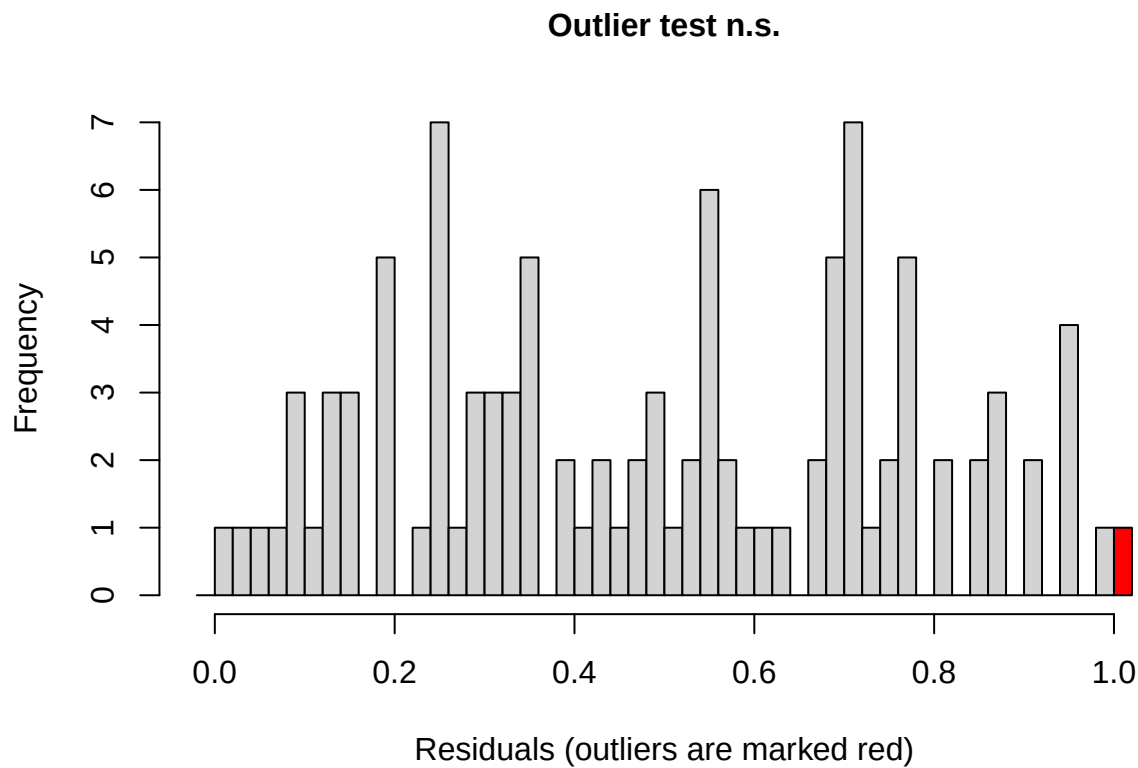
**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**



Simulated values, red line = fitted model. p-value (two.sided) = 0.518

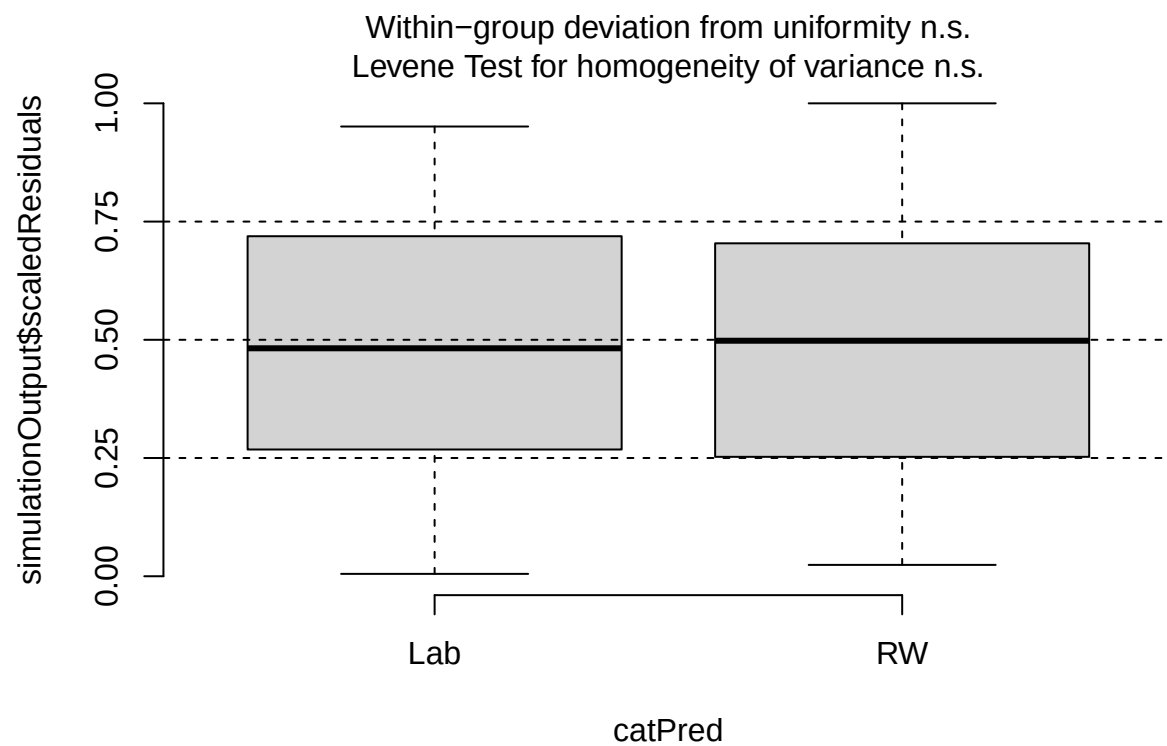
```
##
## DHARMA nonparametric dispersion test via sd of residuals fitted vs.
## simulated
##
## data: simulationOutput
## dispersion = 0.89878, p-value = 0.518
## alternative hypothesis: two.sided
```

```
testOutliers(smr_sim_res)
```

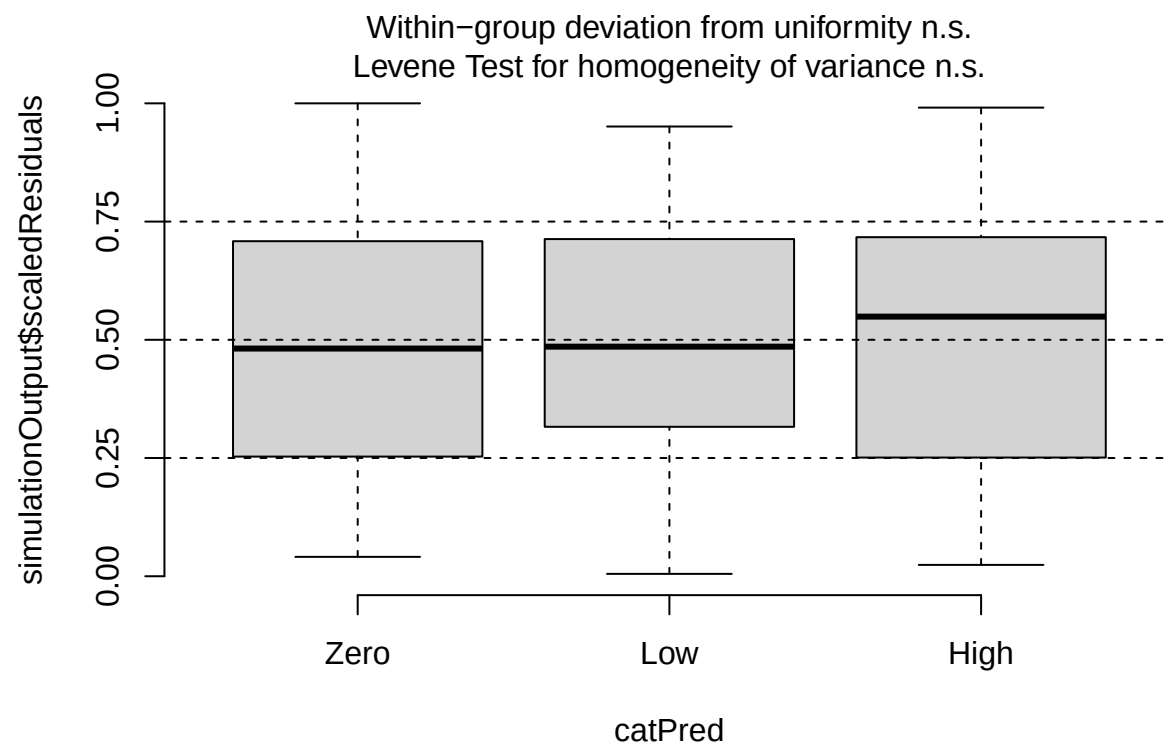


```
##
## DHARMa outlier test based on exact binomial test with approximate
## expectations
##
## data: smr_sim_res
## outliers at both margin(s) = 1, observations = 104, p-value = 0.1878
## alternative hypothesis: true probability of success is not equal to 0.001998002
## 95 percent confidence interval:
## 0.0002434108 0.0524106997
## sample estimates:
## frequency of outliers (expected: 0.001998001998002 )
##                                0.009615385
```

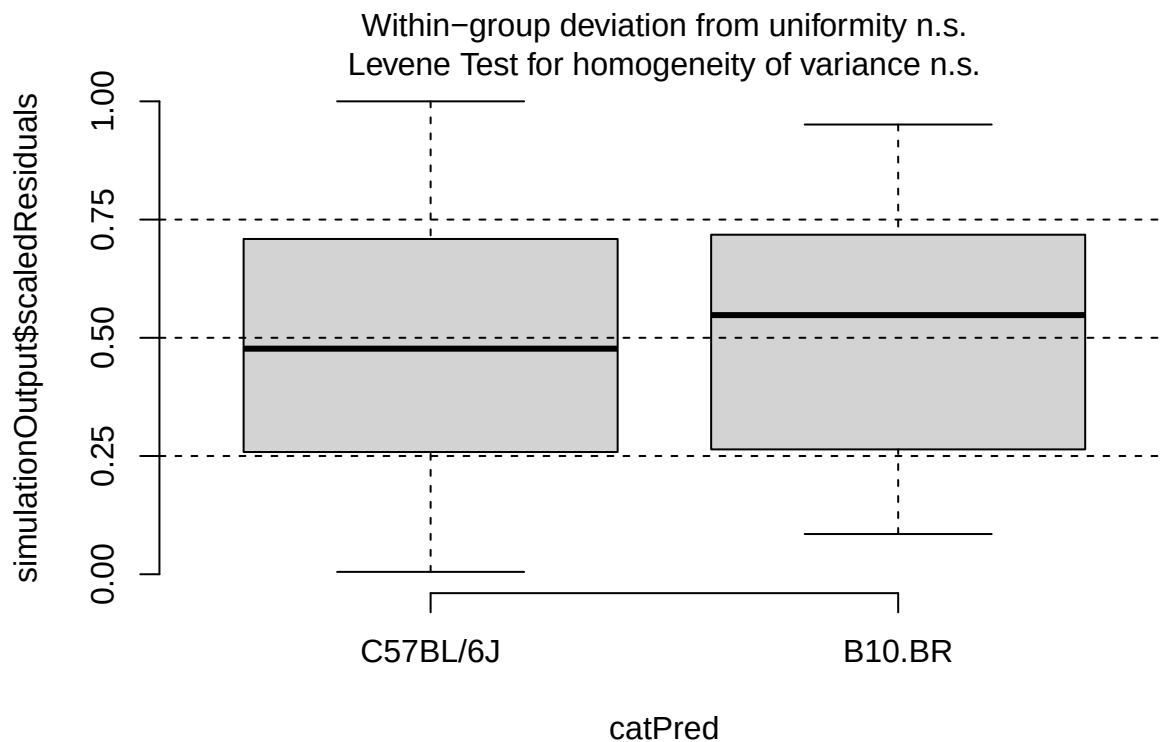
```
plotResiduals(smr_sim_res, form = spleen_mass_compiled_data$lab_or_rw)
```



```
plotResiduals(smr_sim_res, form = spleen_mass_compiled_data$t_muris_dose)
```



```
plotResiduals(smr_sim_res, form = spleen_mass_compiled_data$strain)
```



```
ranef(smr_lmer)
```

```
## $cage_wedge
##      (Intercept)
## 1  -1.250881e-04
## 10  5.543406e-05
## 11  2.945940e-05
## 12  6.333799e-05
## 13 -1.243214e-04
## 14  9.325549e-05
## 15 -4.255035e-05
## 16  8.888178e-05
## 17  2.109006e-05
## 18  1.790580e-05
## 19 -9.216798e-05
## 2   1.295746e-04
## 20  5.012696e-05
## 21  5.234955e-05
## 22  7.523576e-05
## 23 -9.769185e-05
## 24 -3.440721e-05
## 25 -7.706586e-06
## 3   -8.629478e-06
## 4   -2.385667e-05
## 5    2.088278e-05
```

```
## 6 6.746902e-05
## 7 6.657919e-05
## 8 -1.169618e-04
## 9 -1.582010e-04
## W3 3.696003e-04
## W4 4.362037e-05
## W5 -3.696003e-04
## W6 -4.362037e-05
##
## with conditional variances for "cage_wedge"
```

```
VarCorr(smr_lmer)
```

```
## Groups      Name      Std.Dev.
## cage_wedge (Intercept) 0.00030686
## Residual              0.00099303
```

lmer and emm of spleen mass only

```
spleen_lmer <- lmer(spleen_mass_g ~
  strain *
  lab_or_rw *
  t_muris_dose +
  (1 | cage_wedge),
  data = spleen_mass_compiled_data)

summary(spleen_lmer)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: spleen_mass_g ~ strain * lab_or_rw * t_muris_dose + (1 | cage_wedge)
## Data: spleen_mass_compiled_data
##
## REML criterion at convergence: -409.1
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.5310 -0.5088 -0.0574  0.5290  3.9275
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## cage_wedge (Intercept) 5.727e-05 0.007568
## Residual              4.833e-04 0.021985
## Number of obs: 104, groups: cage_wedge, 29
##
## Fixed effects:
##
##              Estimate Std. Error    df
## (Intercept)    0.084411   0.007816 83.035499
## strainB10.BR    0.006489   0.010996 85.175178
## lab_or_rwRW   -0.003235   0.011982 41.579056
```



```

## t_muris_doseLow          0.032399  0.010405 91.798782
## t_muris_doseHigh         0.017650  0.012073 91.628888
## strainB10.BR:lab_or_rwRW -0.004740  0.016908 42.243074
## strainB10.BR:t_muris_doseLow -0.023173  0.014596 90.577539
## strainB10.BR:t_muris_doseHigh  0.012517  0.017527 91.886531
## lab_or_rwRW:t_muris_doseLow -0.004655  0.014686 88.759044
## lab_or_rwRW:t_muris_doseHigh  0.014188  0.015928 91.056604
## strainB10.BR:lab_or_rwRW:t_muris_doseLow  0.012902  0.020698 87.425168
## strainB10.BR:lab_or_rwRW:t_muris_doseHigh -0.028259  0.022745 90.842652
##               t value Pr(>|t|)
## (Intercept)      10.800 < 2e-16 ***
## strainB10.BR       0.590  0.55668
## lab_or_rwRW      -0.270  0.78848
## t_muris_doseLow    3.114  0.00246 **
## t_muris_doseHigh   1.462  0.14717
## strainB10.BR:lab_or_rwRW -0.280  0.78056
## strainB10.BR:t_muris_doseLow -1.588  0.11586
## strainB10.BR:t_muris_doseHigh  0.714  0.47695
## lab_or_rwRW:t_muris_doseLow -0.317  0.75201
## lab_or_rwRW:t_muris_doseHigh  0.891  0.37542
## strainB10.BR:lab_or_rwRW:t_muris_doseLow  0.623  0.53466
## strainB10.BR:lab_or_rwRW:t_muris_doseHigh -1.242  0.21728
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##               (Intr) stB10.BR lb__RW t_mr_L t_mr_H stB10.BR:__RW sB10.BR:__L
## stranB10.BR      -0.711
## lab_or_rwRW      -0.652  0.464
## t_murs_dsLw      -0.710  0.505  0.463
## t_mrs_dsHgh      -0.622  0.442  0.406  0.472
## stB10.BR:__RW    0.462 -0.650 -0.709 -0.328 -0.288
## sB10.BR:__L      0.506 -0.705 -0.330 -0.713 -0.336  0.458
## sB10.BR:__H      0.428 -0.600 -0.279 -0.325 -0.689  0.391  0.455
## lb_r_RW:__L      0.503 -0.358 -0.633 -0.709 -0.334  0.449  0.505
## lb_r_RW:__H      0.471 -0.335 -0.590 -0.358 -0.758  0.418  0.255
## sB10.BR:__RW:__L -0.357  0.497  0.449  0.503  0.237 -0.631 -0.705
## sB10.BR:__RW:__H -0.330  0.463  0.413  0.250  0.531 -0.581 -0.351
##               sB10.BR:__H l__RW:__L l__RW:__H sB10.BR:__RW:__L
## stranB10.BR
## lab_or_rwRW
## t_murs_dsLw
## t_mrs_dsHgh
## stB10.BR:__RW
## sB10.BR:__L
## sB10.BR:__H
## lb_r_RW:__L      0.230
## lb_r_RW:__H      0.522  0.483
## sB10.BR:__RW:__L -0.321 -0.710 -0.343
## sB10.BR:__RW:__H -0.771 -0.338 -0.700  0.476

```

```

emm_spleen_mass <- emmeans(spleen_lmer,
  specs = c("t_muris_dose", "lab_or_rw"),
  by = c("strain"),

```

```

                                type = "response")
emm_spleen_mass

```

```

## strain = C57BL/6J:
##   t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
##   Zero        Lab       0.0844 0.00788 73.9   0.0687   0.100
##   Low         Lab       0.1168 0.00737 81.5   0.1021   0.131
##   High        Lab       0.1021 0.00954 84.2   0.0831   0.121
##   Zero        RW       0.0812 0.00909 12.5   0.0615   0.101
##   Low         RW       0.1089 0.00909 12.5   0.0892   0.129
##   High        RW       0.1130 0.00909 12.5   0.0933   0.133
##
## strain = B10.BR:
##   t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
##   Zero        Lab       0.0909 0.00777 81.6   0.0754   0.106
##   Low         Lab       0.1001 0.00737 80.5   0.0855   0.115
##   High        Lab       0.1211 0.01050 85.1   0.1003   0.142
##   Zero        RW       0.0829 0.00909 12.5   0.0632   0.103
##   Low         RW       0.1004 0.00909 12.5   0.0807   0.120
##   High        RW       0.0990 0.00877 10.9   0.0797   0.118
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95

```

```

pairs(emm_spleen_mass)

```

```

## strain = C57BL/6J:
##   contrast          estimate      SE    df t.ratio p.value
##   Zero Lab - Low Lab -0.032399 0.0105 91.5  -3.082  0.0315
##   Zero Lab - High Lab -0.017650 0.0122 91.2  -1.442  0.7013
##   Zero Lab - Zero RW   0.003235 0.0120 24.5   0.269  0.9998
##   Zero Lab - Low RW   -0.024509 0.0120 24.5  -2.038  0.3511
##   Zero Lab - High RW  -0.028603 0.0120 24.5  -2.378  0.2028
##   Low Lab - High Lab   0.014749 0.0117 90.2   1.255  0.8081
##   Low Lab - Zero RW    0.035635 0.0117 24.4   3.045  0.0548
##   Low Lab - Low RW     0.007890 0.0117 24.4   0.674  0.9832
##   Low Lab - High RW    0.003797 0.0117 24.4   0.324  0.9995
##   High Lab - Zero RW   0.020885 0.0132 32.1   1.585  0.6140
##   High Lab - Low RW   -0.006859 0.0132 32.1  -0.520  0.9950
##   High Lab - High RW  -0.010953 0.0132 32.1  -0.831  0.9594
##   Zero RW - Low RW    -0.027744 0.0104 73.2  -2.677  0.0924
##   Zero RW - High RW   -0.031838 0.0104 74.2  -3.058  0.0352
##   Low RW - High RW    -0.004094 0.0104 74.2  -0.393  0.9987
##
## strain = B10.BR:
##   contrast          estimate      SE    df t.ratio p.value
##   Zero Lab - Low Lab -0.009227 0.0103 83.9  -0.897  0.9462
##   Zero Lab - High Lab -0.030167 0.0129 92.0  -2.345  0.1872
##   Zero Lab - Zero RW   0.007976 0.0120 25.6   0.667  0.9840
##   Zero Lab - Low RW   -0.009498 0.0120 25.6  -0.794  0.9659
##   Zero Lab - High RW  -0.008120 0.0117 23.8  -0.693  0.9810
##   Low Lab - High Lab  -0.020940 0.0124 89.4  -1.684  0.5459

```

```
## Low Lab - Zero RW      0.017203 0.0117 24.2    1.471  0.6851
## Low Lab - Low RW      -0.000272 0.0117 24.2   -0.023  1.0000
## Low Lab - High RW     0.001107 0.0115 22.4    0.097  1.0000
## High Lab - Zero RW    0.038143 0.0139 35.7    2.753  0.0895
## High Lab - Low RW     0.020669 0.0139 35.7    1.492  0.6716
## High Lab - High RW    0.022047 0.0137 34.0    1.615  0.5948
## Zero RW - Low RW     -0.017474 0.0104 74.2   -1.679  0.5500
## Zero RW - High RW    -0.016096 0.0101 73.5   -1.592  0.6067
## Low RW - High RW      0.001378 0.0101 73.5    0.136  1.0000
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 6 estimates
```

```
emm_spleen_mass_df <- as.data.frame(emm_spleen_mass)

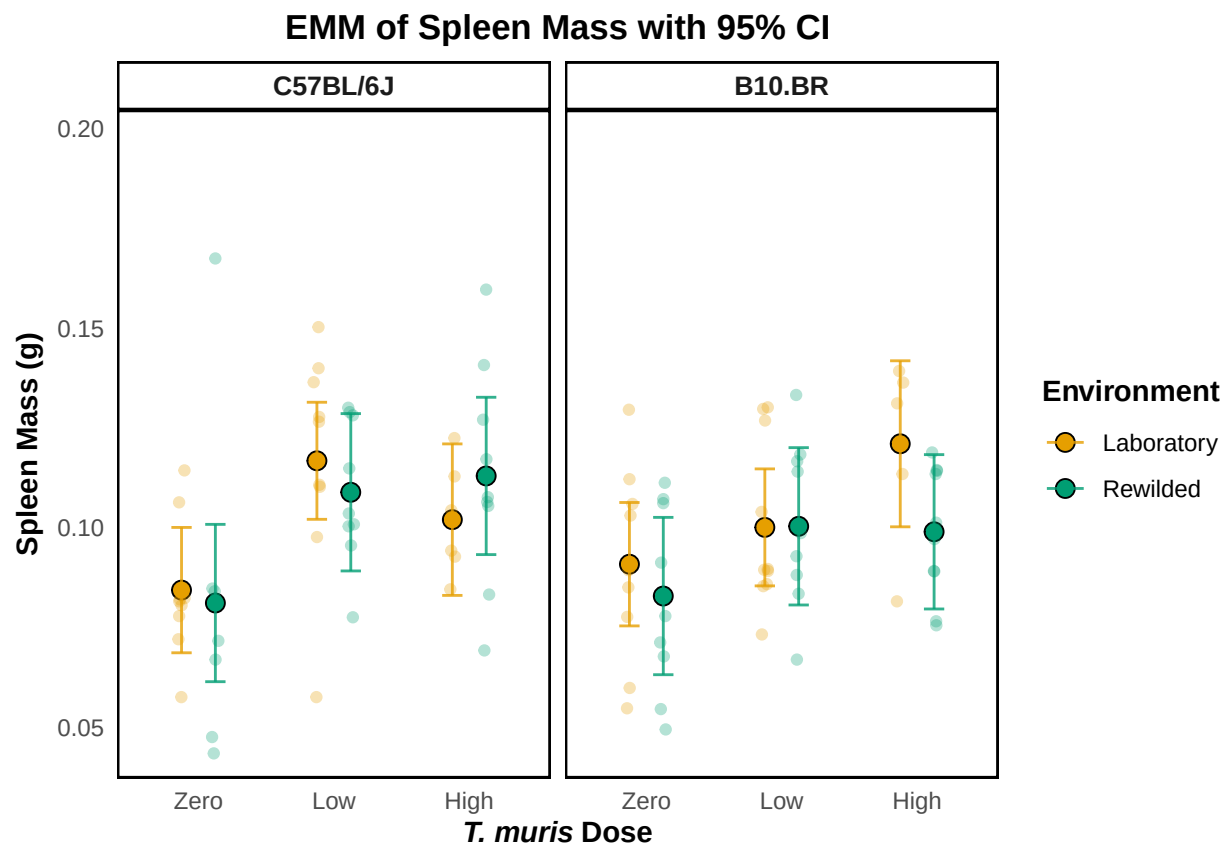
emm_spleen_mass_plot <- ggplot(emm_spleen_mass_df) +
  geom_point(aes(x = t_muris_dose,
                 y = emmean,
                 fill = lab_or_rw,
                 group = lab_or_rw),
             position = position_dodge(width = 0.5),
             alpha = 1,
             size = 3,
             shape = 21,
             color = "black") +
  geom_errorbar(aes(x = t_muris_dose,
                   ymin = lower.CL,
                   ymax = upper.CL,
                   color = lab_or_rw,
                   group = lab_or_rw),
               width = 0.3,
               alpha = 0.8,
               position = position_dodge(width = 0.5)) +
  geom_point(data = spleen_mass_compiled_data,
             aes(x = t_muris_dose,
                 y = spleen_mass_g,
                 color = lab_or_rw),
             position = position_jitterdodge(jitter.width = 0.1,
                                             dodge.width = 0.5),
             alpha = 0.3,
             size = 1.5) +
  scale_y_continuous(name = expression(bold("Spleen Mass (g)")),
                     expand = expansion(mult = c(0.05, 0.3))) +
  scale_x_discrete(name = expression(bolditalic("T. muris")~bold("Dose")))) +
  scale_fill_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                    name = expression(bold("Environment")),
                    labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
  scale_color_manual(values = c("Lab" = color_mustard,
                                "RW" = color_green),
                     name = expression(bold("Environment")),
                     labels = c("Lab" = "Laboratory",
                                "RW" = "Rewilded")) +
```

```

facet_wrap(~ strain) +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                      color = "black",
                                      linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                  fill = NA,
                                  linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
ggtitle("EMM of Spleen Mass with 95% CI")

```

emm_spleen_mass_plot



```

plot_saver("emm_spleen_mass_plot.png",
  emm_spleen_mass_plot)

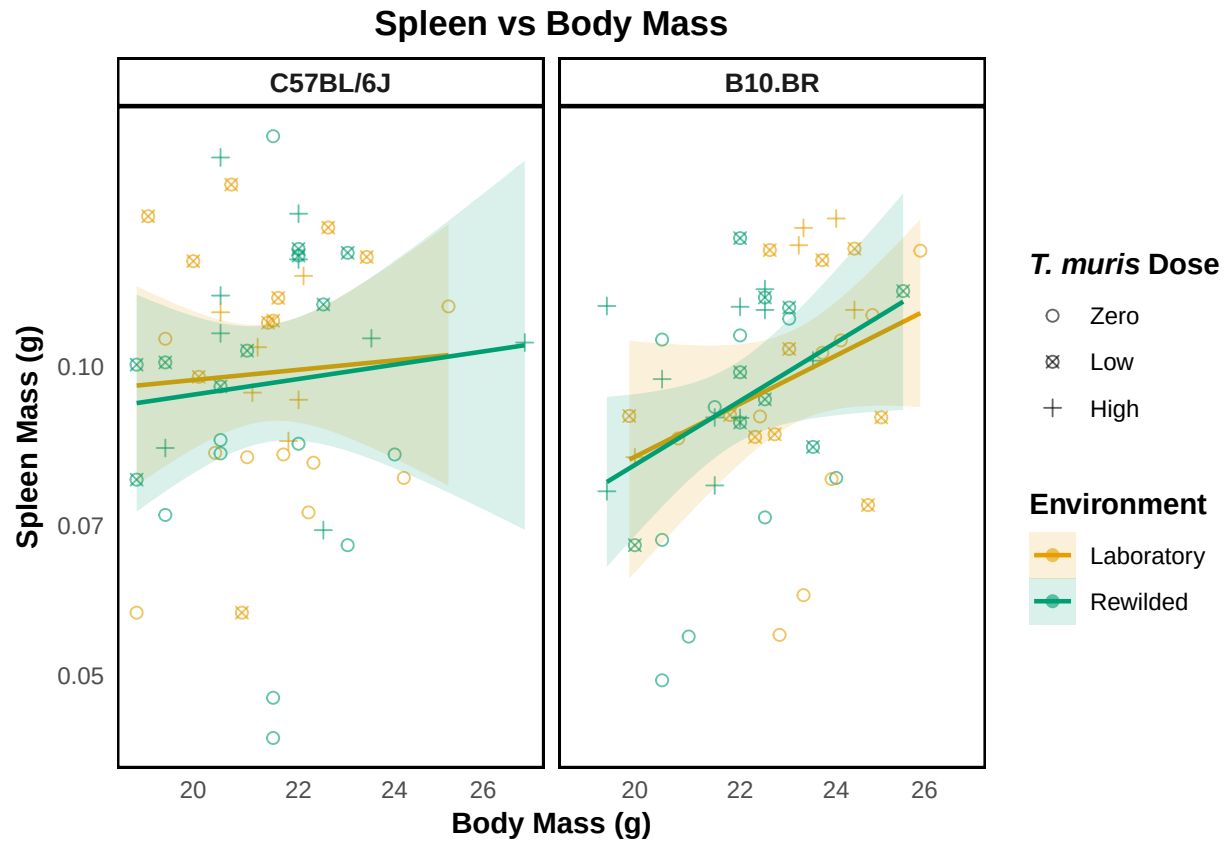
```

We use log scale here to make sure that the whole thing fits.

```
spleen_mass_reg <- ggplot(spleen_mass_compiled_data,
                          aes(x = mass,
                              y = spleen_mass_g,
                              fill = lab_or_rw,
                              color = lab_or_rw,
                              group = lab_or_rw)) +
  geom_point(aes(shape = t_muris_dose),
             alpha = 0.6,
             size = 1.8) +
  geom_smooth(method = "lm",
             se = TRUE,
             alpha = 0.15,
             linewidth = 0.8) +
  facet_wrap(~ strain) +
  scale_fill_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                   name = expression(bold("Environment")),
                   labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
  scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
                    name = expression(bold("Environment")),
                    labels = c("Lab" = "Laboratory",
                              "RW" = "Rewilded")) +
  scale_shape_manual(values = c("Zero" = 1,
                              "Low" = 13,
                              "High" = 3),
                    name = expression(bolditalic("T. muris") ~
                                      bold("Dose"))) +
  scale_x_continuous(trans = "log10",
                    name = expression(bold("Body Mass (g)"))) +
  scale_y_continuous(trans = "log10",
                    name = expression(bold("Spleen Mass (g)"))) +
  theme_minimal() +
  theme(strip.background = element_rect(fill = "white",
                                         color = "black",
                                         linewidth = 0.8),
        strip.text = element_text(size = 10,
                                   face = "bold"),
        panel.border = element_rect(color = "black",
                                     fill = NA,
                                     linewidth = 0.8),
        panel.grid = element_blank(),
        axis.title.x = element_text(face = "bold"),
        axis.title.y = element_text(face = "bold"),
        plot.title = element_text(hjust = 0.5, face = "bold")) +
  ggtitle("Spleen vs Body Mass")

spleen_mass_reg
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
plot_saver("spleen_vs_body_mass_reg.png",
  spleen_mass_reg)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

Spleen mass adjusted for body mass for B6 and B10. I'm splitting the strains

to not overfit the model.

```
spleen_adj_lmer <- lmer(spleen_mass_g ~
  lab_or_rw *
  t_muris_dose *
  strain +
  mass +
  (1 | cage_wedge),
  data = spleen_mass_compiled_data)

summary(spleen_adj_lmer)
```

```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: spleen_mass_g ~ lab_or_rw * t_muris_dose * strain + mass + (1 |
```

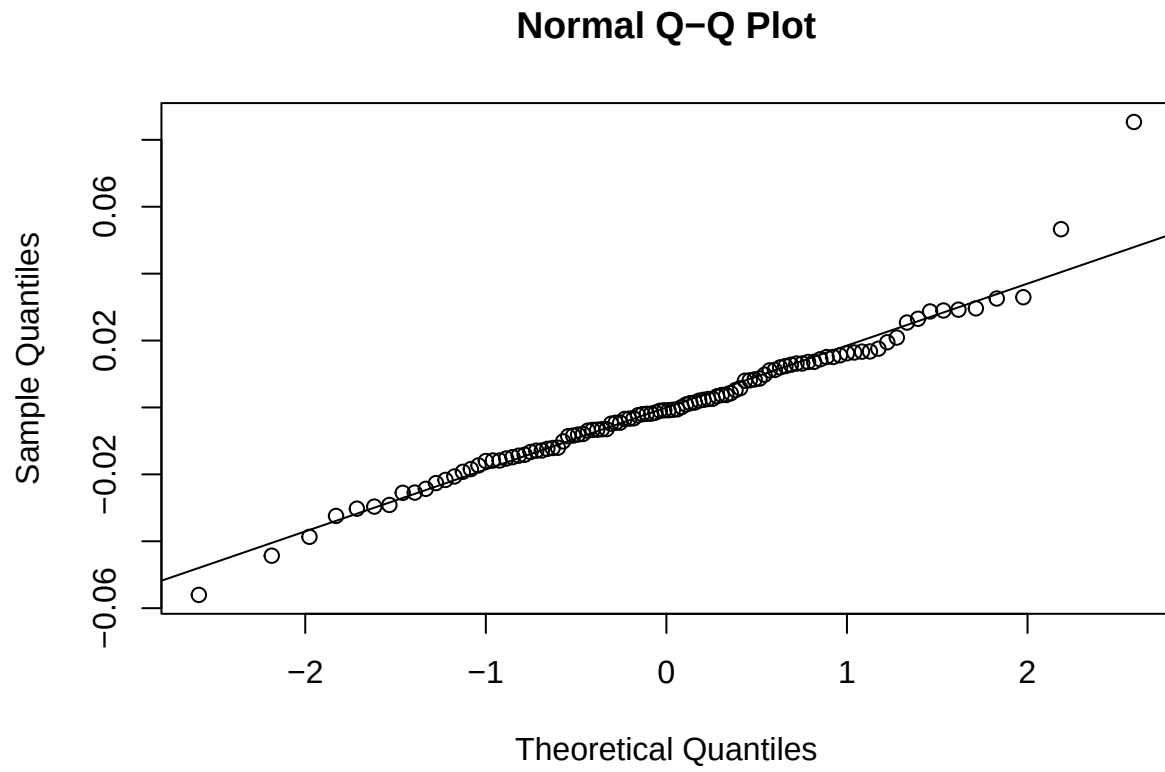
```

##      cage_wedge)
##      Data: spleen_mass_compiled_data
##
## REML criterion at convergence: -403.6
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -2.6025 -0.5809 -0.0391  0.5793  3.9633
##
## Random effects:
##      Groups      Name      Variance Std.Dev.
## cage_wedge (Intercept) 4.772e-05 0.006908
## Residual              4.635e-04 0.021530
## Number of obs: 104, groups: cage_wedge, 29
##
## Fixed effects:
##
##              Estimate Std. Error      df
## (Intercept)      6.004e-03  3.365e-02  9.098e+01
## lab_or_rwRW      -2.368e-03  1.155e-02  4.417e+01
## t_muris_doseLow    3.474e-02  1.020e-02  9.086e+01
## t_muris_doseHigh   1.882e-02  1.178e-02  9.064e+01
## strainB10.BR       2.436e-05  1.103e-02  8.593e+01
## mass              3.603e-03  1.508e-03  9.099e+01
## lab_or_rwRW:t_muris_doseLow -4.797e-03  1.436e-02  8.816e+01
## lab_or_rwRW:t_muris_doseHigh  1.118e-02  1.561e-02  9.029e+01
## lab_or_rwRW:strainB10.BR      8.987e-04  1.646e-02  4.601e+01
## t_muris_doseLow:strainB10.BR -2.343e-02  1.426e-02  8.976e+01
## t_muris_doseHigh:strainB10.BR  1.322e-02  1.709e-02  9.090e+01
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  8.165e-03  2.035e-02  8.709e+01
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -2.621e-02  2.223e-02  9.003e+01
##
##              t value Pr(>|t|)
## (Intercept)      0.178  0.85881
## lab_or_rwRW      -0.205  0.83844
## t_muris_doseLow    3.407  0.00098 ***
## t_muris_doseHigh   1.598  0.11359
## strainB10.BR       0.002  0.99824
## mass              2.390  0.01893 *
## lab_or_rwRW:t_muris_doseLow -0.334  0.73908
## lab_or_rwRW:t_muris_doseHigh  0.716  0.47571
## lab_or_rwRW:strainB10.BR      0.055  0.95671
## t_muris_doseLow:strainB10.BR -1.643  0.10380
## t_muris_doseHigh:strainB10.BR  0.773  0.44134
## lab_or_rwRW:t_muris_doseLow:strainB10.BR  0.401  0.68921
## lab_or_rwRW:t_muris_doseHigh:strainB10.BR -1.179  0.24145
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

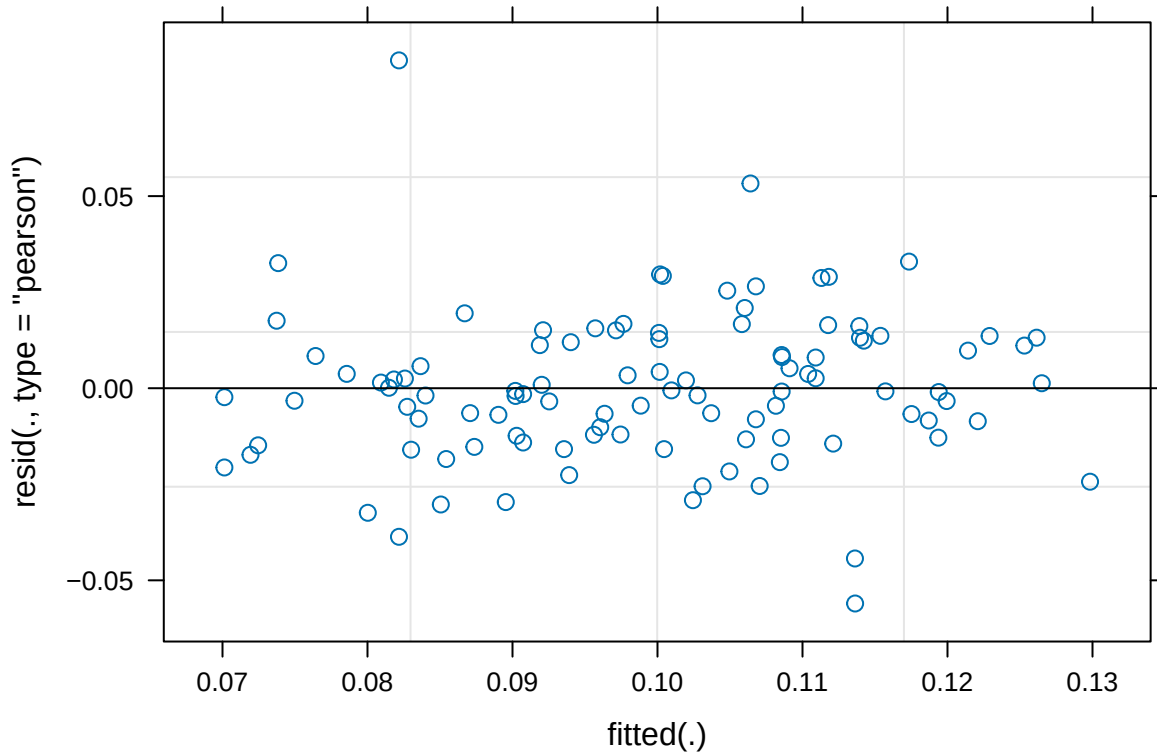
##
## Correlation matrix not shown by default, as p = 13 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it

```

```
qqnorm(resid(spleen_adj_lmer))  
qqline(resid(spleen_adj_lmer))
```



```
plot(spleen_adj_lmer)
```

Let's add the body mass as a co-variate to see if there is any change.

```
emm_adj_spleen <- emmeans(spleen_adj_lmer,
  specs = c("t_muris_dose", "lab_or_rw"),
  by = c("strain"),
  type = "response")

summary(emm_adj_spleen)
```

```
## strain = C57BL/6J:
## t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
## Zero        Lab      0.0852 0.00766 74.5   0.0699   0.100
## Low         Lab      0.1200 0.00729 82.0   0.1054   0.134
## High        Lab      0.1040 0.00932 83.9   0.0855   0.123
## Zero        RW       0.0828 0.00873 12.6   0.0639   0.102
## Low         RW       0.1128 0.00886 13.2   0.0937   0.132
## High        RW       0.1128 0.00870 12.5   0.0940   0.132
##
## strain = B10.BR:
## t_muris_dose lab_or_rw emmean      SE    df lower.CL upper.CL
## Zero        Lab      0.0852 0.00794 81.9   0.0694   0.101
## Low         Lab      0.0965 0.00733 82.5   0.0820   0.111
## High        Lab      0.1173 0.01030 84.4   0.0968   0.138
```

```
## Zero      RW      0.0838 0.00870 12.6    0.0649    0.103
## Low       RW      0.0984 0.00874 12.7    0.0795    0.117
## High      RW      0.1008 0.00841 11.0    0.0822    0.119
##
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
```

```
pairs(emm_adj_spleen)
```

```
## strain = C57BL/6J:
## contrast      estimate      SE    df t.ratio p.value
## Zero Lab - Low Lab -3.47e-02 0.01030 90.7  -3.373  0.0136
## Zero Lab - High Lab -1.88e-02 0.01190 90.1  -1.577  0.6158
## Zero Lab - Zero RW   2.37e-03 0.01160 25.0   0.204  0.9999
## Zero Lab - Low RW   -2.76e-02 0.01170 25.2  -2.365  0.2065
## Zero Lab - High RW  -2.76e-02 0.01160 25.1  -2.383  0.2004
## Low Lab - High Lab   1.59e-02 0.01150 89.3   1.388  0.7344
## Low Lab - Zero RW    3.71e-02 0.01130 25.0   3.289  0.0316
## Low Lab - Low RW     7.17e-03 0.01130 24.9   0.635  0.9871
## Low Lab - High RW    7.12e-03 0.01140 25.3   0.627  0.9879
## High Lab - Zero RW   2.12e-02 0.01270 32.8   1.665  0.5633
## High Lab - Low RW   -8.76e-03 0.01270 32.9  -0.687  0.9821
## High Lab - High RW  -8.81e-03 0.01280 32.8  -0.690  0.9818
## Zero RW - Low RW    -2.99e-02 0.01020 73.1  -2.938  0.0486
## Zero RW - High RW   -3.00e-02 0.01020 74.5  -2.931  0.0493
## Low RW - High RW    -4.71e-05 0.01040 75.4  -0.005  1.0000
##
## strain = B10.BR:
## contrast      estimate      SE    df t.ratio p.value
## Zero Lab - Low Lab -1.13e-02 0.01010 83.8  -1.120  0.8717
## Zero Lab - High Lab -3.20e-02 0.01260 91.0  -2.549  0.1211
## Zero Lab - Zero RW   1.47e-03 0.01190 28.0   0.124  1.0000
## Zero Lab - Low RW   -1.32e-02 0.01160 26.6  -1.136  0.8621
## Zero Lab - High RW  -1.55e-02 0.01170 26.6  -1.325  0.7691
## Low Lab - High Lab  -2.07e-02 0.01210 88.5  -1.707  0.5310
## Low Lab - Zero RW    1.28e-02 0.01140 26.0   1.119  0.8693
## Low Lab - Low RW    -1.90e-03 0.01130 24.9  -0.168  1.0000
## Low Lab - High RW   -4.22e-03 0.01130 24.5  -0.375  0.9989
## High Lab - Zero RW   3.35e-02 0.01350 37.1   2.476  0.1577
## High Lab - Low RW    1.88e-02 0.01340 36.4   1.404  0.7244
## High Lab - High RW   1.65e-02 0.01340 35.6   1.232  0.8182
## Zero RW - Low RW    -1.47e-02 0.01030 74.1  -1.430  0.7090
## Zero RW - High RW   -1.70e-02 0.00991 73.2  -1.715  0.5264
## Low RW - High RW    -2.32e-03 0.01000 73.7  -0.231  0.9999
##
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 6 estimates
```

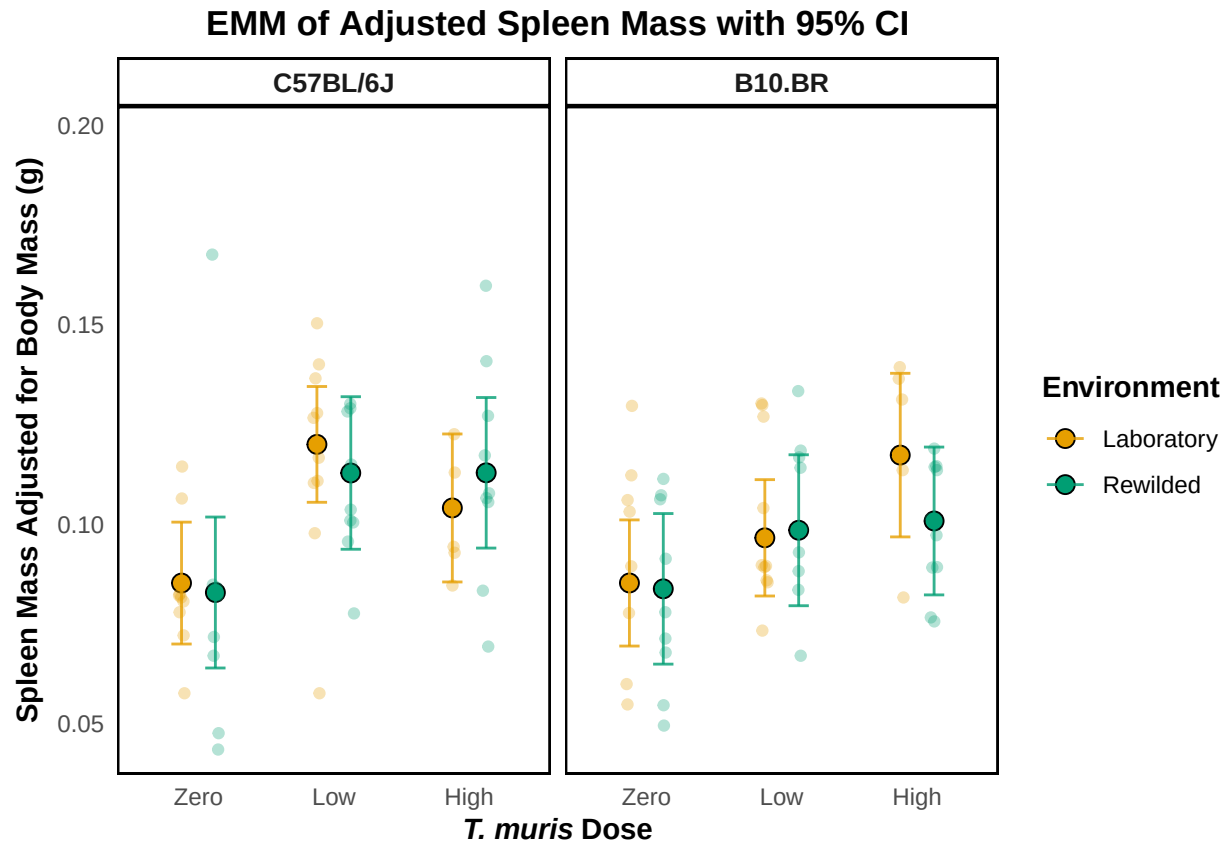
```
emm_adj_spleen_df <- as.data.frame(emm_adj_spleen)
```

```
emm_adj_spleen_plot <- ggplot(emm_adj_spleen_df) +
  geom_point(aes(x = t_muris_dose,
                 y = emmean,
```

```

        fill = lab_or_rw,
        group = lab_or_rw),
    position = position_dodge(width = 0.5),
    alpha = 1,
    size = 3,
    shape = 21,
    color = "black") +
  geom_errorbar(aes(x = t_muris_dose,
    ymin = lower.CL,
    ymax = upper.CL,
    color = lab_or_rw,
    group = lab_or_rw),
    width = 0.3,
    alpha = 0.8,
    position = position_dodge(width = 0.5)) +
  geom_point(data = spleen_mass_compiled_data,
    aes(x = t_muris_dose,
    y = spleen_mass_g,
    color = lab_or_rw),
    position = position_jitterdodge(jitter.width = 0.1,
    dodge.width = 0.5),
    alpha = 0.3,
    size = 1.5) +
  scale_y_continuous(name = expression(bold("Spleen Mass Adjusted for Body Mass (g)")),
    expand = expansion(mult = c(0.05, 0.3))) +
  scale_x_discrete(name = expression(bolditalic("T. muris")~bold("Dose")) +
  scale_fill_manual(values = c("Lab" = color_mustard,
    "RW" = color_green),
    name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
    "RW" = "Rewilded")) +
  scale_color_manual(values = c("Lab" = color_mustard,
    "RW" = color_green),
    name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
    "RW" = "Rewilded")) +
  facet_wrap(~ strain) +
  theme_minimal() +
  theme(strip.background = element_rect(fill = "white",
    color = "black",
    linewidth = 0.8),
    strip.text = element_text(size = 10,
    face = "bold"),
    panel.border = element_rect(color = "black",
    fill = NA,
    linewidth = 0.8),
    panel.grid = element_blank(),
    axis.title.x = element_text(face = "bold"),
    axis.title.y = element_text(face = "bold"),
    plot.title = element_text(hjust = 0.5, face = "bold")) +
  ggtitle("EMM of Adjusted Spleen Mass with 95% CI")
emm_adj_spleen_plot

```



```
plot_saver("emm_adj_spleen_plot.png",
  emm_adj_spleen_plot)
```

```
lym_lmer <- lmer(log(spleen_mass_ratio) ~
  log(total_lym_number) +
  lab_or_rw *
  t_muris_dose +
  (1 | cage_wedge),
  data = lym_number_spleen_data)
```

```
## boundary (singular) fit: see help('isSingular')
```

```
summary(lym_lmer)
```

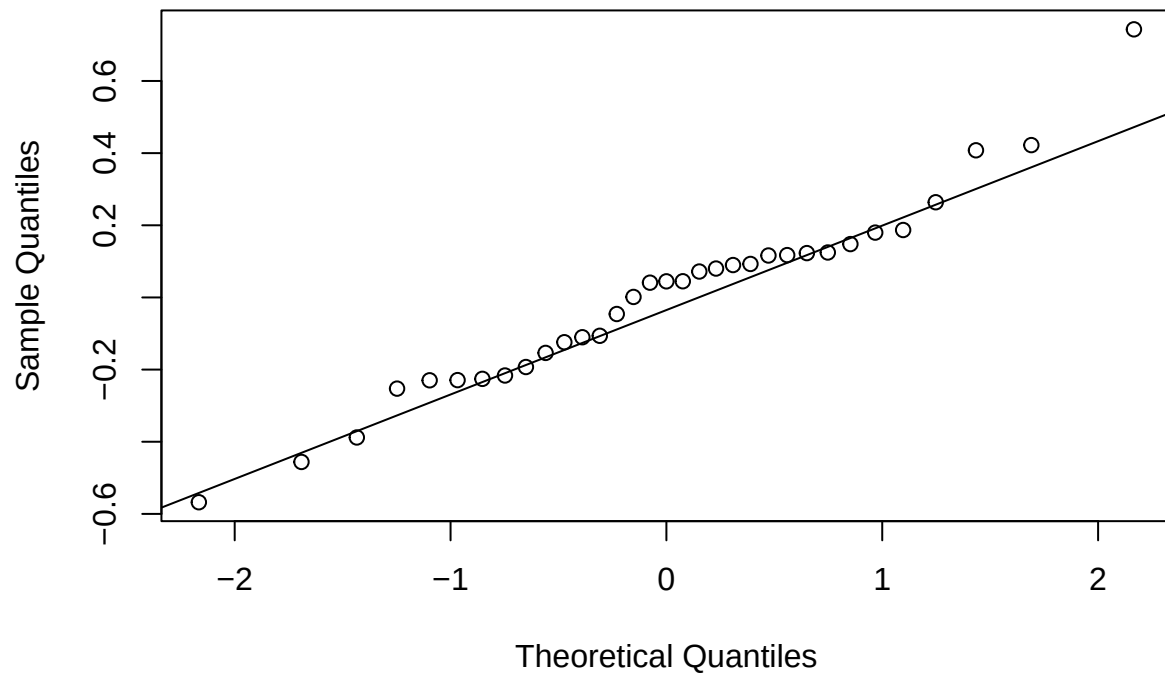
```
## Linear mixed model fit by REML. t-tests use Satterthwaite's method [
## lmerModLmerTest]
## Formula: log(spleen_mass_ratio) ~ log(total_lym_number) + lab_or_rw *
##       t_muris_dose + (1 | cage_wedge)
## Data: lym_number_spleen_data
##
## REML criterion at convergence: 17.5
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -2.0267 -0.6885 0.1591 0.4387 2.6529
##
## Random effects:
## Groups      Name      Variance Std.Dev.
## cage_wedge (Intercept) 0.00000 0.0000
## Residual      0.07845 0.2801
## Number of obs: 33, groups: cage_wedge, 14
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)    -8.10901    3.14072 28.00000   -2.582   0.0154 *
## log(total_lym_number) 0.14303    0.17689 28.00000    0.809   0.4256
## lab_or_rwRW    -0.06095    0.13438 28.00000   -0.454   0.6537
## t_muris_doseHigh 0.16526    0.15933 28.00000    1.037   0.3085
## lab_or_rwRW:t_muris_doseHigh 0.12473    0.19937 28.00000    0.626   0.5366
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) lg(__) lb__RW t_mr_H
## lg(ttl_ly_) -1.000
## lab_or_rwRW -0.206 0.186
## t_mrs_dsHgh 0.359 -0.376 0.337
## lb_r_RW:__H 0.128 -0.114 -0.672 -0.643
## optimizer (nloptwrap) convergence code: 0 (OK)
## boundary (singular) fit: see help('isSingular')
```

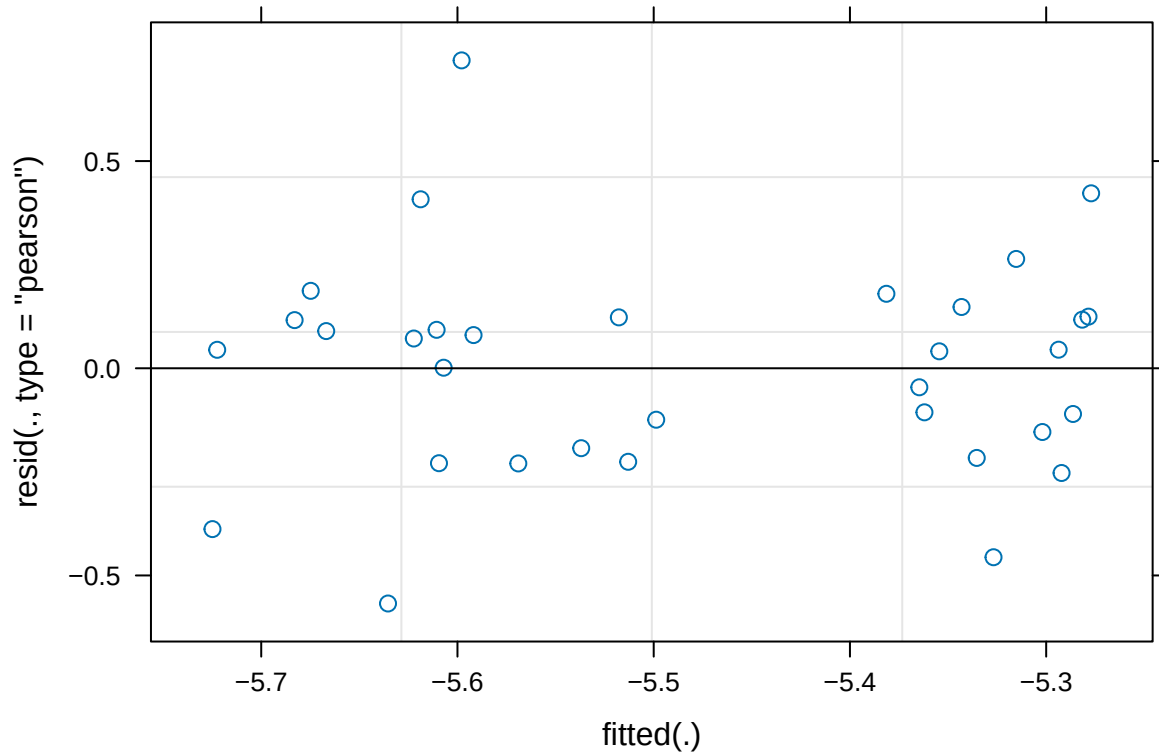
Diagnostic for the lym-mass relationship

```
qqnorm(resid(lym_lmer))
qqline(resid(lym_lmer))
```

Normal Q-Q Plot



```
plot(lym_lmer)
```



Plotting the splenocyte count and spleen mass

```
spleen_mass_lym_count_plot <- ggplot(lym_number_spleen_data,
  aes(x = spleen_mass_g,
    y = total_lym_number,
    fill = lab_or_rw)) +

  geom_point(size = 2,
    alpha = 0.8,
    shape = 21,
    color = "black") +
  geom_smooth(method = "lm",
    se = TRUE,
    fill = "grey70",
    aes(group = lab_or_rw,
      color = lab_or_rw),
    alpha = 0.2) +
  facet_wrap(~ t_muris_dose,
    labeller = as_labeller(c("Zero" = "Zero",
      "High" = "High")))) +
  scale_color_manual(values = c("Lab" = color_mustard,
    "RW" = color_green),
    name = expression(bold("Environment")),
    labels = c("Lab" = "Laboratory",
```

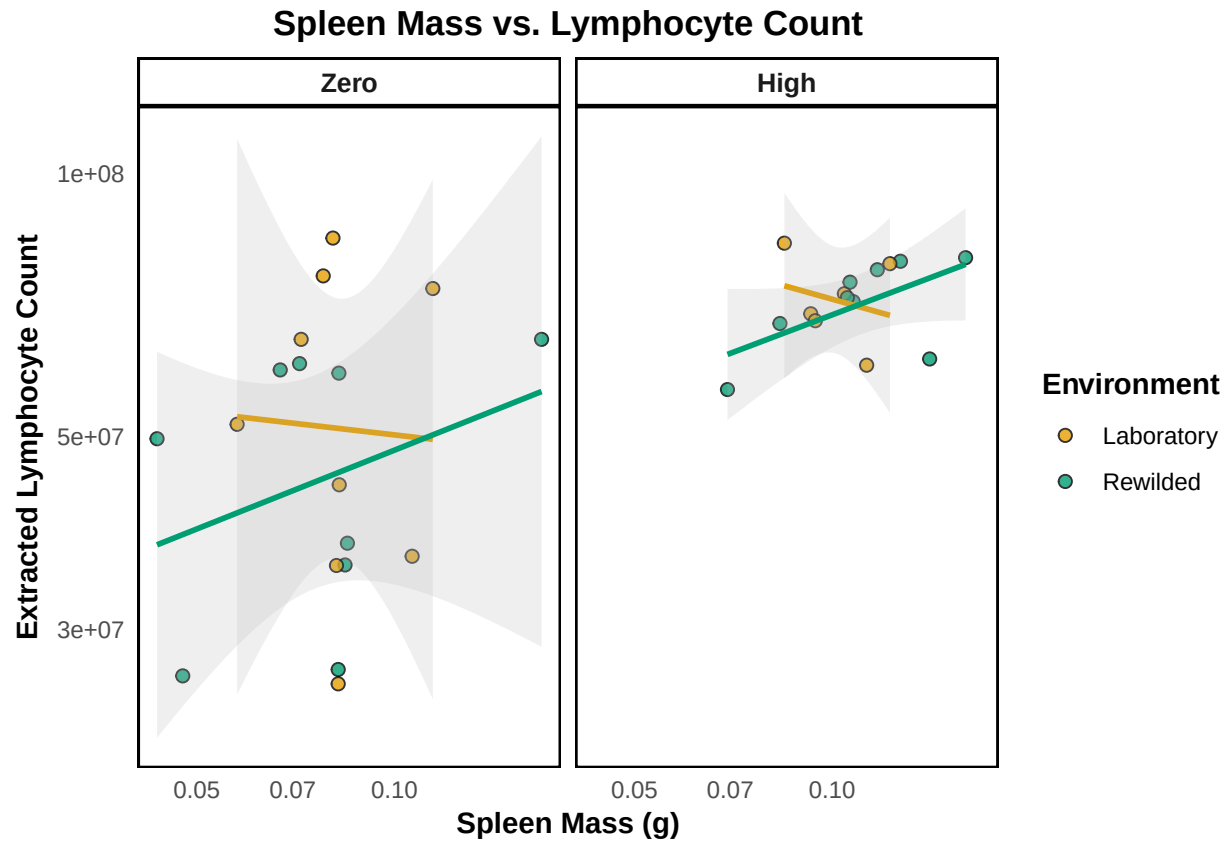
```

      "RW" = "Rewilded")) +
scale_fill_manual(values = c("Lab" = color_mustard,
      "RW" = color_green),
      name = expression(bold("Environment")),
      labels = c("Lab" = "Laboratory",
      "RW" = "Rewilded")) +
scale_x_continuous(trans = "log10",
      name = expression(bold("Spleen Mass (g)"))) +
scale_y_continuous(trans = "log10",
      name = expression(bold("Extracted Lymphocyte Count"))) +
guides(fill = guide_legend(override.aes = list(shape = 21, color = "black")),
      color = "none") +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
      color = "black",
      linewidth = 0.8),
      strip.text = element_text(size = 10, face = "bold"),
      panel.border = element_rect(color = "black",
      fill = NA,
      linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      plot.title = element_text(hjust = 0.5, face = "bold")) +
labs(title = "Spleen Mass vs. Lymphocyte Count")

spleen_mass_lym_count_plot

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

```
plot_saver("spleen_mass_lymphocyte.png",
  spleen_mass_lym_count_plot)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```

CBC lym count and spleen mass at final time point regression

```
spleen_cbc_reg <- ggplot(spleen_cbc,
  aes(x = lym_number,
    y = spleen_mass_g,
    color = lab_or_rw)) +
  geom_point(aes(shape = t_muris_dose),
    size = 2.5,
    alpha = 0.8) +
  geom_smooth(method = "lm",
    se = TRUE,
    aes(group = lab_or_rw,
      fill = lab_or_rw),
    alpha = 0.05,
    linewidth = 0.8,
    show.legend = FALSE) +
  facet_wrap(~ strain) +
```

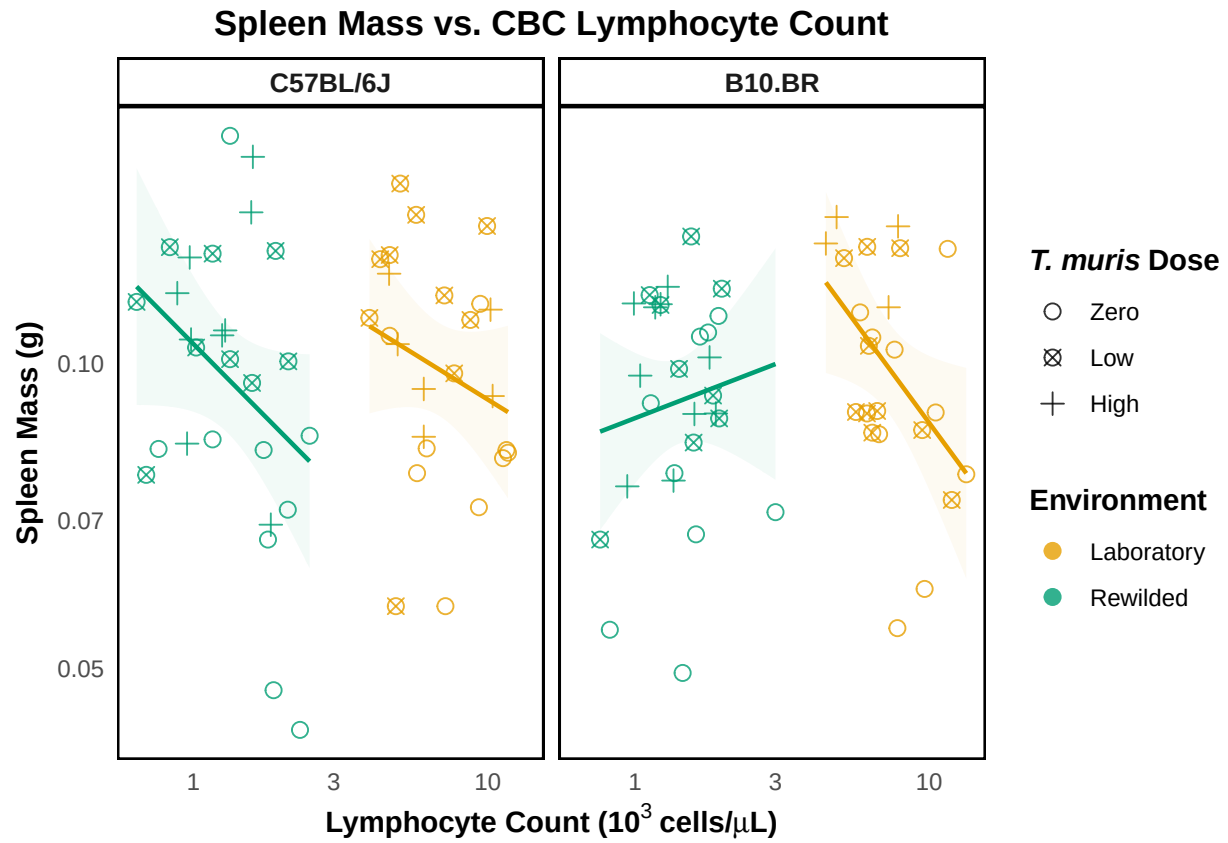
```

scale_color_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
  name = expression(bold("Environment")),
  labels = c("Lab" = "Laboratory",
             "RW" = "Rewilded")) +
scale_fill_manual(values = c("Lab" = color_mustard,
                              "RW" = color_green),
  name = expression(bold("Environment")),
  labels = c("Lab" = "Laboratory",
             "RW" = "Rewilded")) +
scale_shape_manual(values = c("Zero" = 1,
                              "Low" = 13,
                              "High" = 3),
  name = expression(bolditalic("T. muris") ~
                    bold("Dose")) +
scale_x_continuous(name = expression(bold("Lymphocyte Count (103~"cells/"*mu*"L)")),
  trans = "log10") +
scale_y_continuous(name = expression(bold("Spleen Mass (g)")),
  trans = "log10") +
theme_minimal() +
theme(strip.background = element_rect(fill = "white",
                                       color = "black",
                                       linewidth = 0.8),
      strip.text = element_text(size = 10,
                                face = "bold"),
      panel.border = element_rect(color = "black",
                                   fill = NA,
                                   linewidth = 0.8),
      panel.grid = element_blank(),
      axis.title.x = element_text(face = "bold"),
      axis.title.y = element_text(face = "bold"),
      plot.title = element_text(hjust = 0.5,
                                face = "bold")) +
ggtitle("Spleen Mass vs. CBC Lymphocyte Count")

spleen_cbc_reg

```

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
plot_saver("spleen_cbc_reg.png",
  spleen_cbc_reg)
```

```
## 'geom_smooth()' using formula = 'y ~ x'
```